# Beyond the Beneficiary: Old-Age Pensions and the Labor Market Outcomes of Ineligible Adults

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October 16, 2026

**Abstract**

In many low-income countries, younger adults provide substantial financial support and caregiving to elderly household members, yet little is known about how these responsibilities shape their economic behavior. This paper asks whether programs that alleviate these responsibilities affect the labor supply and earnings of working-age adults. Studying Uganda's old-age pension program, I find that an additional year of pension exposure increases monthly sales and profits among ineligible self-employed adults by 10.2 percent and 9.5 percent, respectively, with no significant effect on wage employees' earnings. Weekly hours worked increased by 1.3 percent, with effects concentrated among the self-employed. Gains are also concentrated among women and are largely explained by increased working capital. The effects are driven by pensions received by men rather than women. Overall, old-age pensions relax resource constraints among younger household members, generating meaningful spillovers beyond direct beneficiaries.

*JEL:* H55, J14, J22
Key Words: Old-Age Pension, Earnings, Labor Supply

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# I. INTRODUCTION

In most low-income countries, old-age support is largely provided within the household rather than through formal retirement systems. Because the majority of workers spend their careers in the informal sector, coverage of contributory pensions remains limited, leaving elderly individuals dependent on younger household members for financial support and caregiving.[1] As population aging accelerates, this reliance is expected to intensify, placing increasing pressure on household resources and potentially constraining economic well-being (Harasty and Ostermeier 2020). How this burden shapes the economic behavior of younger individuals, however, remains unclear. This paper asks how programs that alleviate this burden affect younger household members

Prior research has shown that non-contributory pension programs aimed at supporting elderly individuals generate important spillovers within the household, particularly for children's health outcomes (Duflo 2003). Far less is known, however, about their effects on ineligible adults. This gap is important: in settings where households pool resources and jointly determine labor supply, transfers to the elderly may alter the allocation of work and income among younger adults. By relaxing household resource constraints and potentially shifting intra-household incentives, pensions may affect labor supply, earnings, and other labor market outcomes of working-age household members. Understanding these spillover effects is central to evaluating the broader welfare implications and cost–benefit of such programs.

Old-age pensions may affect the labor market outcomes of younger individuals through several channels. First, pension income reduces the financial dependence of the elderly on younger household members. For instance, Fan (2010) show that pensions decrease the likelihood that adult children transfer resources to their parents. While often interpreted as crowding out of private transfers, this also reflects a decline in the elderly's reliance on younger individuals, potentially freeing up resources for the latte

Second, pensions relax household budget constraints more broadly. Evidence shows that beneficiaries allocate part of the transfer to health, nutrition (Duflo 2003), and children's education (Moscona and Seck 2024). To the extent that these expenditures are partly financed by younger adults in the absence of pensions, this reallocation may release resources that can be directed toward labor market activities. However, the resulting income effect may also reduce labor supply, rendering the overall impact on work incentives theoretically ambiguous.

Third, younger household members, particularly those who are self-employed, may

[1] Throughout the paper, "younger individuals" refers to pension-ineligible adults aged 18 to 64.

act as intermediaries for investing pension income in productive activities, thereby affecting earnings and labor supply. Finally, pensions may alter the intra-household allocation of time. Existing evidence shows that pension receipt leads beneficiaries to reduce labor supply or exit the labor market (Fetter and Lockwood 2018; Unnikrishnan and Imai 2020). If the elderly reallocate this time toward household production and caregiving, younger adults, especially women, may face lower time constraints and increase their labor market participation. Consistent with this mechanism, Ardington et al. (2009) show that pensions relax childcare constraints for prime-aged adults, facilitating migration for work.

The effects of old age pensions on the earnings and labor supply of younger household members are likely to differ from those of direct cash transfers to this group, operating through distinct channels. To the extent that younger individuals invest pension resources in productive activities, the presence of the elderly beneficiary introduces a layer of social accountability, and potentially oversight, that is absent when transfers are received directly, which may discipline how the resources are deployed. Alternatively, if the effects operate through expenditure substitution, whereby pension income finances household consumption previously borne by younger members, the resulting reduction in the financial dependence of the elderly frees up resources for younger adults without any direct transfer changing hands. In either case, the pension alters the intra-household allocation of resources in ways that a direct transfer to younger adults would not replicate.

To address the question of interest, I exploit an unusual policy experiment in Uganda: the pilot of the Senior Citizen Grant (SCG), an old-age pension program introduced in 15 districts in 2011. In the pilot districts, the program provided a monthly grant of $9.8 (at the 2012 exchange rate) to all individuals aged 65 years and above, and to those aged 60 years and above in the Karamoja sub-region. At baseline, the transfer amount was estimated to represent roughly 20% of average monthly per-capita consumption.

I use difference-in-differences design to estimate the causal effect of the pension program on the labor market outcomes of younger household. Following Moscona and Seck (2024), I construct a measure of potential pension exposure based on the number of household members eligible for the pension at any point in time. Cumulatively, this measure captures the total years of pension benefits a household would have received had it been located in the pilot districts. By construction, it varies across households depending on the age composition of their members. I combine this household-level variation with geographic variation induced by the rollout of the program across pilot and non-pilot districts. The analysis draws on data from the Uganda National Household Survey and

the Demographic and Health Survey. To examine labor supply responses more closely, I also estimate dynamic effects using a staggered difference-in-differences framework, where treatment timing is defined by the eligibility of the oldest household member. This complementary analysis uses data from the Uganda National Panel Surveys.

The results show that exposure to old age pensions positively affects the earnings and labor supply of ineligible adults in the household. One additional year of pension exposure raises the monthly sales and profits of self-employed individuals by 10.2 percent and 9.5 percent, respectively, corresponding to $11.7 in sales and $4.8 in profits relative to the mean in non-pilot districts. These effects are especially pronounced among women, with sales and profits increasing by 18.0 percent and 17.4 percent, respectively. No significant effects are detected on the earnings of wage employees. Pension exposure also raises labor supply: an additional year increases weekly hours worked by 1.3 percent in the full sample and by 1.4 percent among women. Consistent with the earnings results, these effects are concentrated among the self-employed

The evidence suggests that these impacts are largely driven by an increase in working capital among self-employed individuals, which rises by 6.4 percent, equivalent to $2.3, indicating that pension income either relaxes financial constraints or provides resources that younger household members channel into productive investment. These effects are economically meaningful given that one year of pension benefits amounts to approximately $118, based on monthly transfers of $9.8.

The impact of pension exposure is amplified by the availability of microcredit and stronger informal networks, suggesting that cash transfers complement rather than substitute for existing financial institutions. Although not the primary focus of the paper, I also show that pensions reduce the labor supply of elderly recipients, consistent with previous studies (e.g., Fetter and Lockwood 2018; Unnikrishnan and Imai 2020). The positive effects on ineligible household members suggest that, in aggregate, the decline in the labor supply of the elderly may be partly offset by increases in the earnings and labor supply of younger adults, a consideration that is relevant for evaluating the net labor market impact of pension programs.

I further examine whether the gender of the pension recipient shapes the effects on the earnings and labor supply of younger adults. I construct separate measures of potential pension exposure based on whether the recipient is male or female, and estimate the effects using the same research design as before. Pension exposure through male recipients has a significant effect on the earnings and labor supply of younger household members, whereas exposure through female recipients does not. Following earlier literature (e.g., Duflo 2003; Unnikrishnan and Imai 2020), I also examine whether the gender

of the recipient matters for children's outcomes. Consistent with prior findings, exposure through female rather than male recipients has a significant impact on children's education and health. To the extent that the effects of pensions on earnings and labor supply are driven by investment of the pension grants, these contrasting patterns, male recipients driving adult labor market effects and female recipients driving children's outcomes, reveal gender differences in preferences: male recipients are more likely to channel transfers toward income-generating activities, whereas female recipients prioritize investments in children's human capital.

The results are robust to several sensitivity checks. They are unchanged when excluding Kampala from the sample, indicating that the findings are not driven by factors specific to the capital city such as differences in labor market structure or access to services. They are also unaffected by controlling for remittances, ruling out the possibility that changes in remittance flows to households with elderly members confound the estimated effects. As a further validation, I conduct a placebo test by constructing a measure of potential pension exposure based on individuals just below the eligibility age cutoff. Since these individuals do not qualify for the pension, exposure through them should have no effect on household outcomes. Consistent with this prediction, the placebo measure yields no significant effects on either earnings or labor supply, reinforcing the interpretation that the main results are driven by actual pension receipt rather than age composition per se.

I also examine the validity of the identifying assumptions through several complementary tests. First, I estimate event-study specifications to test for differential pre-trends in pilot districts. I implement a cohort-based analysis, grouping households by their distance from eligibility for those not yet age-eligible and by the number of years since eligibility for those already covered. The coefficients for cohorts not yet eligible are indistinguishable from zero, supporting the parallel trends assumption. By contrast, coefficients for eligible cohorts are positive and significant, consistent with a treatment effect that increases with exposure. Second, I exploit multiple waves of the Uganda National Panel Survey to test for pre-trends in labor supply outcomes directly.[2] Using a staggered difference-in-differences design following De Chaisemartin and d'Haultfoeuille (2020), I find no evidence that the estimated effects on labor supply are driven by pre-existing trends. Finally, I show that the findings are not explained by endogenous household formation, i.e., younger adults selectively migrating into households with pension-eligible members, which would confound the interpretation of the spillover effects.

[2] A similar analysis cannot be conducted for earnings, as the panel does not report earnings for self-employed individuals.

This paper contributes to the literature on the impacts of pensions on household income (Huang and Zhang 2021) and on the labor supply of direct beneficiaries (Fetter and Lockwood 2018; Unnikrishnan and Imai 2020) by examining how old-age pensions affect the earnings and labor supply of ineligible adults. Causal identification of these spillover effects has been difficult because most pension programs are universal in design, leaving eligibility criteria as the primary source of variation. This study exploits an unusual policy experiment in Uganda, where the staggered rollout of an old-age pension program generated variation along both eligibility and geographic dimensions, allowing for a cleaner identification of impacts on ineligible household members. Data constraints have also limited prior research on earnings effects. For instance, Huang and Zhang (2021) show that China's pension program increased household income, but their analysis does not disentangle whether this increase was driven solely by direct transfers or also by spillovers to younger individuals' activities. This paper differs by identifying the specific sources of household income growth, drawing on a unique module in the 2016/17 UNHS that collected detailed earnings information, including sales and profits of self-employed individuals. The findings show that old-age pensions have a multiplier effect by raising the earnings of self-employed ineligible adults. With respect to spillover effects on labor supply, this study is closely related to Ardington et al. (2009), who show that large transfers to the elderly in South Africa increased employment among prime-aged adults, primarily through labor migration. Their analysis, however, focuses on the extensive margin, whereas this paper examines both extensive and intensive margins and finds stronger responses along the latter. In addition, unlike Ardington et al. (2009), the focus here is on individuals residing within the same household as the pension beneficiary.

This paper also relates to a large literature on the effects of pension programs on the welfare of beneficiaries and their households. Existing studies show that these programs improve the health and well-being of recipients (Huang and Zhang 2021; Miglino et al. 2023; Alzua et al. 2023), increase household consumption (Fan 2010; Unnikrishnan and Imai 2020), and raise non-food expenditures and asset accumulation (Unnikrishnan and Imai 2020). There is also evidence of important spillovers to other household members, including improvements in children's health and nutrition (Duflo 2003). Building on this literature, I provide evidence that old-age pension programs also affect the welfare of ineligible adults within the household by increasing their earnings.

More broadly, the paper contributes to the debate on whether the gender of the recipient shapes the effects of pension and other cash transfer programs. Prior work finds that transfers received by women tend to have larger impacts on children's health (Duflo

2003) and household consumption (Unnikrishnan and Imai 2020) than those received by men. I show that the role of recipient gender depends on the outcomes considered: while pensions received by women have stronger effects on children's outcomes, those received by men have larger effects on the earnings and labor supply of ineligible adults.

Finally, this paper contributes to the literature on the effects of unconditional cash transfers. Using experimental and quasi-experimental methods, studies have documented positive effects of such transfers on children's educational and health outcomes (Akee et al. 2010; Baird et al. 2014; Shah and Gennetian 2023), household consumption and psychological well-being (Haushofer and Shapiro 2016), and health-seeking behaviors (Novignon et al. 2022), alongside negative effects on labor market participation (Verlaat et al. 2023). However, the transfers studied in this literature are typically short-term or one-off in nature. This paper examines a distinct form of unconditional transfer, an old age pension, that generates a permanent and predictable increase in household income, potentially giving rise to different behavioral responses than those observed in temporary transfer programs.

The rest of the paper is organized as follows. Section II describes the Senior Citizen Grant. Section III describes the data sources used in the study, and presents the description of the sample. Section IV describes the identification strategy for the paper. Section V presents the main empirical results and their discussion. Section VI presents the conclusion.

## II. THE SENIOR CITIZEN GRANT IN UGANDA

The Senior Citizen Grant (SCG) is an unconditional cash transfer program in Uganda which targets individuals aged 65 and above (60 and above for Karamoja sub-region).[3] The program was adopted in 2011 with the main goal of reducing old-age poverty by providing a minimum level of income security to the beneficiaries. It is implemented as part of the broader Expanding Social Protection (ESP) program designed in 2009 with the aim of "reducing chronic poverty and improving life chances for poor men, women and children by embedding a national social protection system within Uganda's national planning and budgeting" (Bukuluki and Watson 2012). The SCG provides a monthly cash transfer of about $9.8 (at the baseline) to the beneficiaries, which is about 20% of average monthly per-capita consumption at the baseline.

Beneficiary selection is based on a registration process by community-based local government structures, involving Local Council I (LC I) leaders. Where possible, the informa-

[3]The age of eligibility is lower in Karamoja because of the region is more economically deprived

tion from local government structures is verified using national civil registry (using national identification data). The program excludes older persons with government-funded pensions as part of their formal retirement package.

The implementation of the program followed a phased approach. Rather than introduce the program in all districts, the government piloted the program in 15 districts for five years—henceforth, Phase I.[4] The pilot phase was launched in March 2011 with the first payout in September 2011. After the completion of the pilot phase in 2015, the government decided to expand the program to an additional 40 districts over the next 5 years, starting with 20 districts in the financial year 2015/16, thereafter to 5 districts every year, until 2019/20—henceforth, Phase II. By the financial year 2016/17, SCG had reached 47 districts—7 of these were new districts carved out of the original pilot districts. In 2017, the Parliament of Uganda passed a resolution to roll out the grants to all districts in the country but this has not yet been fully actualized due to financial constraints.[5] Figure 1 shows the staggered roll-out of the program.

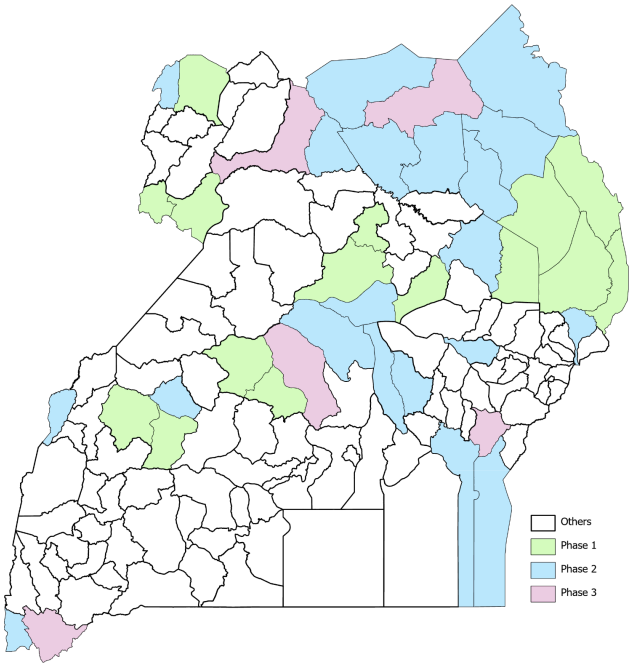



Figure 1: Map of Uganda showing the roll-out of the Senior Citizen Grant

*Note:* Figure 1 shows the staggered roll-out of the Senior Citizen Grant in different districts in Uganda. The unshaded districts are the not-yet treated districts—where the program has not yet reached as of today. The different colors indicate the different phases of the program. Note that here, Phase 2 and 3 are broadly categorized as Phase II in the rest of the paper since phase 3 happens just one year after phase 2 and both were under the additional 40 districts earlier mentioned.

[4]It's not clear what criteria was followed in selecting the pilot districts

[5]Information about the roll out was obtained from the ESP website. https://socialprotection.go.ug. Accessed February 2024

Phase II came with some administrative and design changes compared to Phase I. First, in Phase I, monthly transfers were sent directly to beneficiaries' mobile money accounts through MTN Mobile Money, and converted into cash by MTN agents at designated pay points. In Phase II, the mode of payment was changed from mobile money to Post Bank Uganda. Under this new arrangement, the bank organizes payout days and delivers the money at designated pay points. Secondly, in Phase II, the program enrolls only 100 oldest persons in each sub-county in the selected districts, while in Phase I all individuals who are aged 65 years and above (or 60 and above in Karamoja) were enrolled in the program and this criteria continues.

The above changes in the design in Phase II make it hard to cleanly define treatment status for households in the selected districts since not all age-eligible individuals in the household were included in the program. One way to circumvent this would be to use administrative data of beneficiaries to define treatment status and then link it to household survey data. This is not possible because I do not have overlapping identifiers in administrative data and household data. For Phase I, it is possible to define treatment status for households in the pilot districts since all age-eligible individuals were included in the program. In this case, the treatment depends on whether the household has age-eligible individuals and the number of such individuals in the household. For this reason, my identification is based on Phase I (pilot districts) and not-yet treated districts. Since my main data set was collected in 2016/17, some Phase II districts had received treatment for at least 1 year. For this reason, I exclude the Phase II districts from the control group in the main analysis. I, however, show that my results are still robust if I use all districts, but slight lower in magnitude, as expected since districts treated in Phase II would bias the effects downwards.

## III. DATA AND DESCRIPTIVE STATISTICS

The main dataset for the paper is the Uganda National Household Survey (UNHS) collected in 2016/17. This is a cross-sectional wave which is the sixth in the series of the national household surveys conducted by Uganda Bureau of Statistics (UBOS). The survey is geographically representative and covers all the districts in Uganda. To account for seasonality, the data collection is spread over a 12 months period. I account for this in my analysis by including month of interview fixed effects. The survey administered three key modules; the socioeconomic, the labor force, and the community modules. This paper uses information from the first two modules. The socioeconomic module collects detailed information on household and individual demographic characteristics, includ-

ing the age of all members. I use information on age composition from this module to define potential exposure to the pension program as discussed in section IV. The labor force module collects detailed information about employment status, job characteristics, hours worked, different measures of unemployment, and household chores (care activities). Unlike the earlier waves and other household surveys, the 2016/17 wave separately classified incomes for self-employed individuals, by collecting information on the sales, operating expenses, and profits. This unusual feature of the data is very important for my paper.[6] It allows me to explore how the earnings of self-employed individuals are affected by the old-age pension.

The other dataset that I use in this study is the Uganda National Panel Surveys (UNPS) 2009/10-2015/16. The UNPS is also a nationally representative household survey which is implemented under the World Bank's Living Standards Measurement Study (LSMS). It's designed to provide statistical information for monitoring changes and transitions in poverty dynamics, trends and related welfare indicators. The survey is carried out over a twelve-month period on a selected sample of households. I leverage the panel structure of this data to test for pre-trends in labor supply and examine alternative mechanisms which might evolve as a result of the pension program, such as endogenous formation of households. This also serves as robustness check for my results from the UNHS data. I do not use the UNPS for my main results albeit its advantages of the panel structure, for the following reasons. First, the panel is limited in terms of sample size which is compounded by the sample refresh in 2013 where one-third of the sample was replaced. This sample size limitation makes it infeasible to disaggregate the analysis by gender and status in employment. Secondly, the panel does not collect detailed information on the earnings of individuals in self-employment, which is also a major aspect of my paper. For these two reasons, I only use the panel for robustness checks and take the results from UNHS as my main results. Lastly, I use the Demographic and Health Surveys (DHS) for Uganda, collected in 2016 to study outcomes on children's education and health.

Table 1 shows the characteristics of the sample (aged 18 to 64 years). In this table, I summarize the mean values for potential pension exposure, individual characteristics, and household characteristics for pilot and non-pilot districts.[7] The average potential pension exposure is 1.4 years. Notably, the potential pension exposure measure is fairly balanced across pilot and non-pilot districts, suggesting absence of migration of elderly individuals from non-pilot to pilot districts. Figure 2 presents the distribution of the po-

[6] In this paper I define self-employed to include employers and own-account workers

[7] For the purpose of the discussion I will refer to non-pilot as being non-pilot districts excluding Phase II districts i.e column 3 in Table 1.

tential pension exposure for pilot and non-pilot districts. It shows that the distribution is similar for pilot and non-pilot, both including individuals with zero potential exposure (Figure 2a), and excluding them (Figure 2a). Figure A.2 in the Appendix also shows that the aggregate potential exposure across the districts is as good as randomly assigned. This suggests that my instrument for pension exposure can be taken to be exogenous. In Table 1, I also show the proportion of individuals potentially exposed to at least one pension year. The table shows that 17% of the individuals in my analysis sample are potentially exposed to at least one pension year. This proportion is slightly higher in pilot districts (20%).

The individual level characteristics show that, on average, individuals work for 36 hours, with pilot districts working about 9.5 hours less compared to non-pilot. Similarly, pilot districts have a lower percentage of individuals engaged in paid employment. While 39% of the individuals in the full sample are engaged in paid employment, only 36% report being engaged in paid employment in pilot districts, and 41% report so in non-pilot districts. The table also shows that individuals in the pilot districts are more likely to be engaged in farm work and skilled agriculture, and less likely to be engaged in elementary occupations, and service and sales work, compared to those in non-pilot. Individuals in pilot districts are less likely to be employees but more likely to be self-employed. Regarding earnings, the average monthly wage in the sample is estimated at $94, with pilot districts having a lower average ($78) compared to non-pilot ($96). Earnings in self-employment (sales and profits) follow a similar pattern as wages. While the differences between pilot and non-pilot are apparent, the gap narrows when I exclude Kampala from the sample, suggesting the differences are partly inflated by the capital city. Accordingly, I conduct a robustness test where I exclude Kampala from the sample.

On individual demographic characteristics, Table 1 shows that the sample is gender balanced across pilot and non-pilot. There is also no significant difference in the average age of individuals across pilot and non-pilot districts, but the pilot districts have a lower mean for years of schooling compared to non-pilot districts. Household characteristics show differences in residence, education of household head, and poverty status. Households in the pilot districts are less likely to be located in urban areas, their household heads have lower levels of education on average, and are more likely to be poor. There is, however, no difference in household size, age, and sex of the household head. Again, some of the differences are accounted for by the capital city.

The point of Table 1 is not to claim balance between pilot and non-pilot districts but rather to get a sense of the sample characteristics and distribution of potential pension exposure across pilot and non-pilot districts. Indeed, some of the differences in Table 1

Table 1: Descriptive statistics: UNHS 2016/17

| | (1) All | (2) Pilot | (3) Non-Pilot (Excl.Phase II) | (4) Non-Pilot (Excl. Kla) | (5) Diff: (2)-(3) |
|---|---|---|---|---|---|
| ***Potential Exposure*** | | | | | |
| Potential Pension Exposure | 1.37 | 1.46 | 1.32 | 1.33 | 0.14 (0.073) |
| $\mathbb{I}\{\text{Potential Pension Exposure} > 0\}$ | 0.17 | 0.20 | 0.17 | 0.17 | 0.03 (0.007) |
| ***Individual characteristics*** | | | | | |
| Hours worked (weekly) | 36.14 | 28.50 | 38.08 | 36.96 | -9.58 (0.435) |
| Paid work | 0.39 | 0.36 | 0.41 | 0.39 | -0.05 (0.009) |
| Farm work | 0.48 | 0.59 | 0.45 | 0.48 | 0.14 (0.010) |
| Skilled agriculture | 0.53 | 0.70 | 0.48 | 0.51 | 0.22 (0.010) |
| Elementary occupations | 0.17 | 0.11 | 0.18 | 0.18 | -0.07 (0.008) |
| Service and sales workers | 0.16 | 0.10 | 0.18 | 0.17 | -0.08 (0.008) |
| Employee in main job | 0.23 | 0.15 | 0.26 | 0.24 | -0.11 (0.009) |
| Self-Employed in main job | 0.47 | 0.54 | 0.45 | 0.45 | 0.09 (0.010) |
| Monthly wage (USD) | 93.73 | 78.20 | 95.93 | 88.74 | -17.73 (6.549) |
| Sales (USD) | 91.44 | 49.97 | 107.61 | 99.27 | -57.64 (4.458) |
| Profits (USD) | 42.07 | 23.08 | 48.51 | 45.65 | -25.43 (1.850) |
| Female | 0.54 | 0.54 | 0.54 | 0.54 | 0.00 (0.010) |
| Age | 33.26 | 33.23 | 33.28 | 33.48 | -0.05 (0.231) |
| Years of schooling | 7.45 | 6.28 | 7.74 | 7.49 | -1.47 (0.084) |
| ***Household characteristics*** | | | | | |
| Household size | 4.52 | 4.73 | 4.38 | 4.46 | 0.36 (0.066) |
| Urban residence | 0.33 | 0.18 | 0.38 | 0.32 | -0.20 (0.012) |
| Age of the head | 42.60 | 42.93 | 42.56 | 43.03 | 0.37 (0.412) |
| Years of schooling of the head | 6.32 | 4.81 | 6.75 | 6.50 | -1.94 (0.113) |
| Poor | 0.22 | 0.29 | 0.19 | 0.20 | 0.11 (0.010) |
| Male headed | 0.69 | 0.67 | 0.69 | 0.69 | -0.02 (0.012) |

*Notes:* This table presents the averages of the samples. The sales and profits are self-reported by self-employed individuals (employers and own-account workers). All monetary values are converted into US dollars using end period exchange rate of 2016/17: \$1 =UGX 3590.90. All the employment related indicators are only based on the main job. All individual level characteristics and statistics for potential pension exposure are computed on a sample of individuals aged 18 to 64. Column 5 presents the difference in the means for pilot and non-pilot (excl. Phase II). The standard errors are presented in parenthesis.

could be attributed to exposure to the pension. To get a better sense of how pilot and non-pilot districts are comparable, I conduct a test of the differences in means at the baseline (in 2011) using the UNPS data. The results in Table 2 show that the potential exposure at the baseline is slightly higher in pilot districts. Note that the magnitudes here are smaller compared to Table 1 because this is potential exposure only in 2011 rather than cumulative exposure. The table also shows that pilot districts had lower labor supply, were more likely to be self-employed in the secondary job, earned lower wages, and had

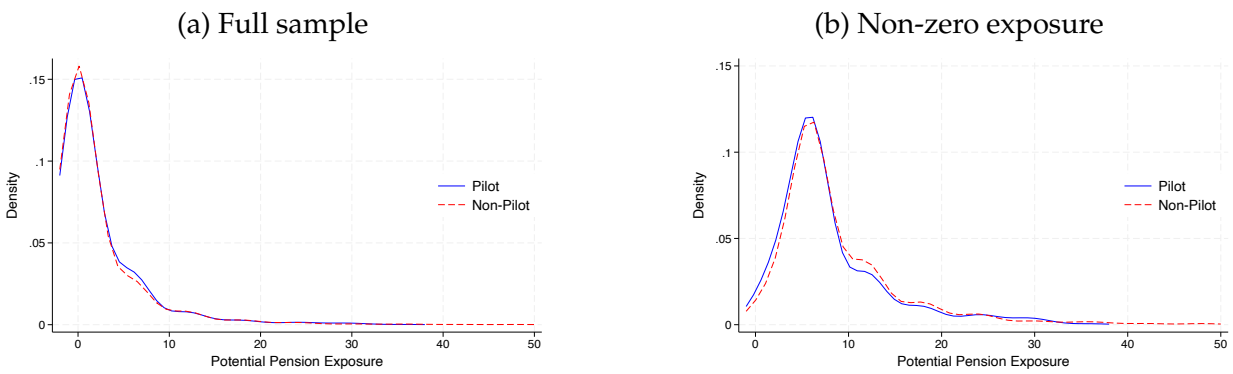



Figure 2: Distribution of potential exposure across pilot and non-pilot districts.

*Notes:* The figure presents the distribution of potential pension exposure for pilot and non-pilot (excl. Phase II) districts. The measure is computed using the formula in equation 1. Figure 2a presents the distribution while including those in the full sample. Figure 2b presents the distribution while excluding those with zero potential exposures.

lower average years of schooling. While these baseline differences are apparent, when I cluster the standard errors at district level (Table A.1 in Appendix), only differences in wages are significant. This suggests that the standard errors are highly correlated within districts.

## IV. IDENTIFICATION

I use quasi-experimental variation in the roll-out of the Senior Citizen Grant (SCG) in Uganda. My main identification exploits the variation in exposure to the program across households and across districts. Following Moscona and Seck (2024), I use the age composition of the household members to construct a measure for Potential Pension Exposure (PPE) which measures the cumulative number of years of pension grants the household would have received if it was located in pension districts. By construction, this measure varies at household level depending on whether the household has pension-eligible individuals and the number of such individuals. To construct this measure, I first take a count of all pension-eligible individuals in the household at any given point in time and then sum up across time, from the time of the program launch to the time of my data. This measure is constructed as shown below.

$$PPE_h = \sum_{t=2011}^{2016} \left( \sum_{i \in h} \mathbb{I}\{Age_{iht} \geq 65\} \right), \tag{1}$$

Table 2: Differences in means for pilot and non-pilot districts in 2011

| | Mean in Non-Pilot | Diff. (Pilot-Non-Pilot) | $N$ |
|---|---|---|---|
| Potential Pension Exposure | 0.223 | 0.051*** | 11810 |
| | (0.005) | (0.014) | |
| Hours worked (weekly) | 24.966 | -3.800*** | 6582 |
| | (0.263) | (0.731) | |
| Self-employed in main job | 0.114 | -0.014 | 7406 |
| | (0.004) | (0.011) | |
| Self-employed in secondary job | 0.349 | 0.082* | 965 |
| | (0.017) | (0.039) | |
| Paid work | 0.140 | -0.009 | 11809 |
| | (0.003) | (0.010) | |
| Monthly wage (USD) | 89.104 | -27.125* | 606 |
| | (3.231) | (12.128) | |
| Female | 0.514 | 0.003 | 11810 |
| | (0.005) | (0.014) | |
| Age | 24.291 | 0.315 | 11778 |
| | (0.177) | (0.501) | |
| Years of schooling | 5.060 | -0.805*** | 10252 |
| | (0.041) | (0.123) | |

*Notes:* This table presents mean differences between pilot and non-pilot (excl. phase II) districts. Each row is a regression of the named variable on a constant and the indicator for pilot. Standard errors are presented in parenthesis. *** p<0.01, ** p< 0.05, * p<0.10.

where $Age_{iht}$ is the age of individual $i$ in household $h$ at time $t$. One assumption I make is that the household composition is not endogenously changing. As I show later, this assumption is reasonable in the setting that I study.

The second source of variation in exposure to the program comes from the staggered roll-out of the pension program across districts. I combine the variation in potential exposure and the geographical variation in the allocation of the program using Difference-in-Difference design. The main regression specification is as follows:

$$y_{ihd} = \beta_0 + \beta_1 \left(PPE_h \cdot \mathbb{I}\{Pilot_d\}\right) + \beta_2 PPE_h + \beta_3 Pilot_d + \gamma' X_{ihd} + \tau_m + \tau_r + e_{ihd}, \quad (2)$$

where $y_{ihd}$ is the outcome variable of individual $i$, living in household $h$, located in district $d$, in this case labor supply and earnings. $Pilot_d$ is the indicator variable for being located in pilot district, $X_{ihd}$ is a vector of control variables including age, sex, education, location (urban/rural), marital status, education of the household head, household size, and employment status of the household head. $\tau_m$ and $\tau_r$ are month of interview and region

fixed effects respectively. In the analysis, I cluster the standard errors at district level to account for potential correlation within a district.

My identification comes from comparing outcomes of individuals living in different households with otherwise similar levels of potential pension exposure but located in different districts (pilot versus non-pilot). Given the controls in the model, the identifying assumption is that individuals in households with the same level of potential pension exposure, same age, same level of education, same sex, same location (rural-urban), same region, interviewed in the same month, living in households with similar characteristics are on similar trends absent the pension program. Under the assumption of exogeneity of the pension instrument (Potential Pension Exposure), I interpret $\beta_1$ as a causal effect of the pension program.

With the approach described above, it's difficult to test for parallel trends assumption, which is crucial in difference-in-difference design. To deal with this issue and other robustness issues, I extend the analysis by leveraging the panel data earlier described. For the panel analysis, I begin by defining the treatment (access to pension) to follow a staggered design. As indicated in section II, the design and implementation of the pension program was such that only individuals aged 65 years and above (60 years for Karamoja) were eligible for the pension grant. This implies households without eligible individuals did not access the program until the oldest person crossed the eligibility threshold. This means households where the oldest person was above 64 years at the time of introduction of the program (in 2011) and were located in pilot districts accessed the program at the onset in 2011. Households whose oldest person was 64 years, are treated one year later, and those whose oldest person was 63 years in 2011 are treated 2 years later, and so on. This introduces staggered adoption among households, which I exploit. For the purposes of the estimation, I make a simplifying assumption that keeps the treatment stable. In practice, the intensity of treatment can increase as more people in the household cross the age eligibility threshold, or decrease if the household loses the pension beneficiary. I assume that once the household gets treated, it doesn't lose the treatment. This is necessary for implementing the staggered DID design since it requires that the treatment is in absorbing state. Because of these assumptions and the caveats of the panel data highlighted earlier, I do not explicitly interpret the magnitudes from this exercise but rather use it to study dynamic trends in labor supply.

To estimate the average treatment effect from the staggered treatment described above, I adopt the Difference-in-Difference estimator ($DID_M$) recently proposed by De Chaisemartin and d'Haultfoeuille (2020). The estimator has been suggested to mitigate the bias arising from negative weights and heterogeneous treatment effects that are not ac-

counted for in earlier estimators such as the Two-Way Fixed Effects (De Chaisemartin and d'Haultfoeuille 2020). The estimator is also the most applicable in cases where, for each pair of consecutive dates, there are groups whose treatment does not change, which is likely the case in my data. I therefore estimate the following regression specification.

$$y_{it} = \alpha_i + \gamma_t + \sum_{\substack{t=2009 \\ t \neq 2011}}^{2015} \beta_t \mathbb{I}\{Pension_{it} = 1\} * \mathbb{I}\{Wave = t\} + \varepsilon_{it}, \tag{3}$$

where $pension_{it}$ is an indicator if the household has a pension eligible individual, $Wave$ indicates the year of observation, $\alpha_i$ is individual fixed effect and $\gamma_t$ is time fixed effect. $\epsilon_{it}$ is the error term.

## V. EMPIRICAL RESULTS AND DISCUSSION

### *V.A. Old-Age Pension and Earnings of Younger Individuals*

Figure 3 shows the summary of the results from equation 2 using sales, profits, and wages as the dependent variables. Table A.2 presents the results in detail. I present the results separately for the full sample, women's sample, and men's sample. The results show that exposure to old-age pension leads to an increase in earnings of younger individuals, but only for those engaged in self-employment. Specifically, Figure 3 shows that, in the full sample, one additional year of pension exposure increases the monthly sales and profits by 10.2% and 9.5%, respectively. Based on the mean in the non-pilot districts, the effect on sales is equivalent to $11.7, while the effect on profits is equivalent to $4.8. These effects are economically meaningful given that one year of pension exposure is equivalent to $118. I do not find significant effect of the pension exposure on earnings of employees (wages) in the full sample, even though the direction suggests negative effects. The effects on sales and profits are only significant among women and substantially higher for this group. I find that one additional year of pension exposure increases sales and profits by 18.0% and 17.4%, respectively, among women. Among men, the effect on sales and profits is positive but not significant. Indeed, Table A.3 in the Appendix shows that the effect of the pension on women is statistically different compared to men.

The positive effect of the old-age pension on earnings is in line with the recent findings by Huang and Zhang (2021) who show that the pension program for seniors in China, the New Rural Pension Scheme (NRPS) significantly increased household income of eligible households in rural China. They show that household income increases by 18% in the full sample but the effect drops to 10% when they exclude the very poor seniors.

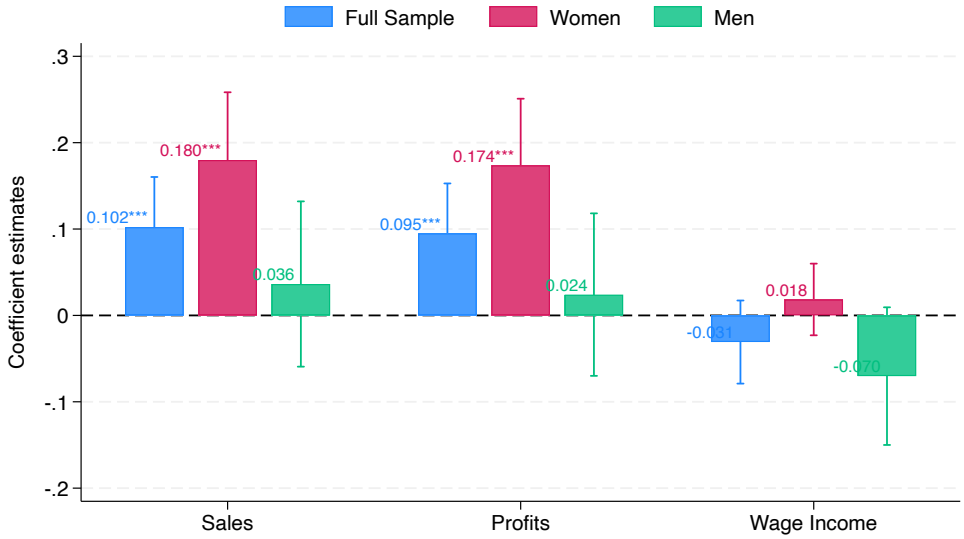



Figure 3: Effects of old-age pension on earnings of ineligible adults

*Notes:* The variables at the bottom are the dependent variables, transformed using Inverse Hyperbolic Sine Transformation (IHS). Each of the estimates is from a different regression. All the models are estimated on a sample of individuals aged 18 to 64. All models include control variables (including interaction terms with PPE), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

*Placebo Experiment:* As a validation exercise, I conduct a placebo experiment where I construct a placebo potential pension exposure based on the individuals that are just below the cutoff age (just below 65 years). In this case, my placebo exposure is based on individuals aged 50 to 64 years, as shown in equation 4, where $PPE_h^p$ is the *Placebo Potential Pension Exposure*.

$$PPE_h^p = \sum_{t=2011}^{2016} \sum_{i \in h} \mathbb{I}\{50 \leq Age_{iht} \leq 64\}, \tag{4}$$

With this placebo measure, I re-estimate the regression in equation 2 including this placebo. Specifically, I estimate the following regression:

$$\begin{aligned} y_{ihd} =& \beta_0 + \beta_1 \left(PPE_h \cdot Pilot_d\right) + \beta_2 \left(PPE_h^p \cdot Pilot_d\right) + \beta_3 PPE_h + \beta_4 PPE_h^p + \\ & \beta_5 Pilot_d + \gamma' X_{ihd} + \tau_m + \tau_r + e_{ihd}, \end{aligned} \tag{5}$$

where $\beta_1$ is the coefficient of the true exposure while $\beta_2$ is the coefficient of the placebo exposure. All the variables are as defined before.

If the original measure of pension exposure is capturing the effects of the pension program, I do not expect the placebo exposure to have the same effects, since individuals

just below the cutoff are not eligible and therefore do not receive the pension. Figure 4 presents $\beta_1$ and $\beta_2$ from the regression in equation 5. The results show that only the true exposure predicts positive and significant effects on sales and profits both in the full sample and women's sample. The coefficients for the placebo exposure are very small in magnitude and not significant. This suggests that the pension instrument is indeed capturing the effects of the pension and not anything else associated with household structure. For robustness, I adjust the interval for the placebo from 50 - 64 to 55 - 64 years. Results in Table A.4 show a similar pattern where only the true exposure produces the true effects.

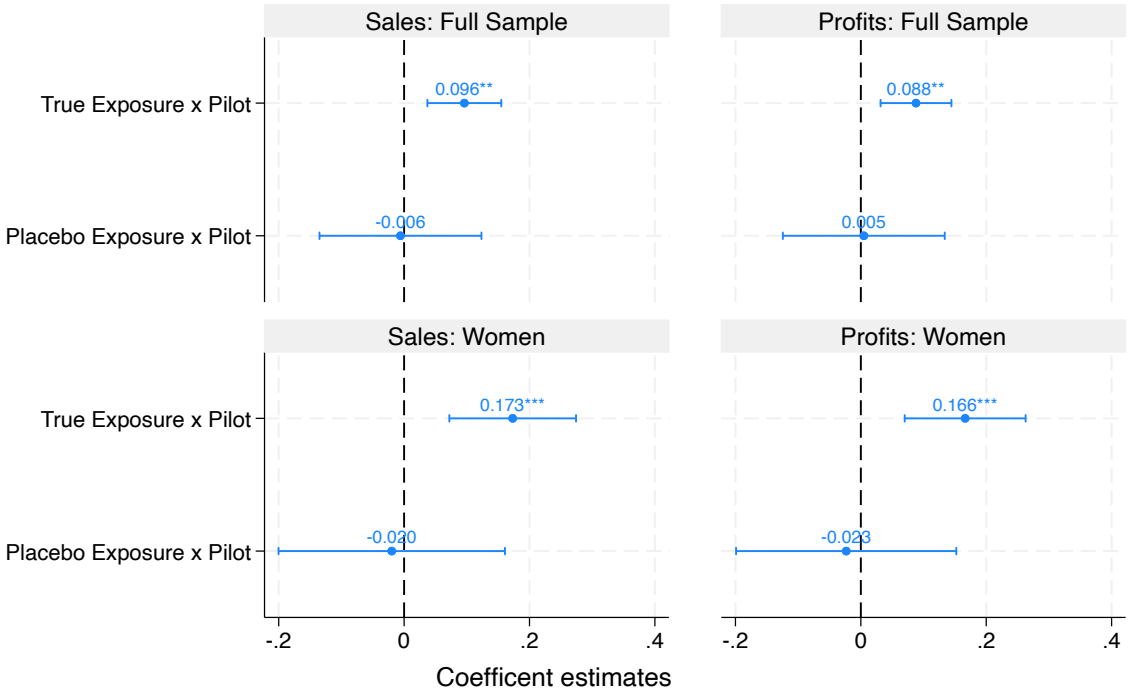



Figure 4: Effects of old-age pension on earnings of ineligible adults: Placebo Experiment

*Notes:* The dependent variables are sales and profits, transformed using inverse hyperbolic sine transformation. The top-right and top-left plots are for the full sample while the bottom-left and bottom-right plots are for the women's sample. True exposure is constructed from equation 1 while placebo exposure is constructed from equation 4. The coefficients are estimated from regression equation 5. All the models are estimated on a sample of individuals aged 18 to 64. All models include control variables (including interaction terms with exposure measures), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

*Event Study Graphs:* To examine if individuals in pilot districts are experiencing differential trends, I conduct a cohort-based analysis where I group households in cohorts based on far away from the pension (for those below eligibility age) and the number of years the household has had the pension. I begin by constructing a measure of the distance to the pension in years which captures the difference between the age of the oldest

person in the household and the eligibility cutoff age of 64. Since the pension program had existed for five years prior to my data, I restrict the highest number of years the household has spent with the pension to five. Equation 6 shows how the Distance to Pension (DTP) is constructed, where $i$ indexes individuals and $h$ indexes households. I then discretize this measure into cohorts of two-year intervals (I do this for power reasons) and estimate cohort-specific effects using regression in equation 7.

$$DTP_h = max_i \left\{ \min\{\text{Age}_{ih} - 64, 5\} \right\}, \tag{6}$$

$$y_{ihc} = \beta_0 + \sum_c \beta_c \left( \mathbb{I}_d^{Pilot} * \mathbb{I}_h^c \right) + \beta_p Pilot_d + \gamma' X_{ihd} + \tau_c + \tau_m + \tau_r + e_{ihd}, \tag{7}$$

where $\sum_c$ sums over two-year cohort bins. $\mathbb{I}_d^{Pilot}$ is an indicator of being located in the pilot district, $\mathbb{I}_h^c$ is an indicator for the household's $DTP_h$ falling in cohort $c$, and $\tau_c$ are cohort fixed effects. $X_{ihd}$ contains all the variables defined before but also includes interactions with cohorts. $\beta_c$ are the coefficients of interest. It captures the differential effect on individuals living in households located in pilot districts by cohort.

Figure 5 presents the coefficients of $\beta_c$ on sales and profits. As expected, $\beta_c$ is indistinguishable from 0 for cohorts that have not yet reached the pension, indicating that pre-trends are parallel. For cohorts that have reached the pension, the coefficients are positive and significant. Note that I do not explicitly interpret the magnitudes here since this analysis doesn't account for the intensity of the treatment arising from the fact that some households may have more than one pension eligible individual. Nonetheless, the coefficients are close in magnitude to those in Figure 3.

The results on earnings beg a question as to what explains the positive effects of old-age pension on the earnings of self-employed individuals. In what follows, I rationalize two interrelated potential channels. First, I test the labor supply channel—does the pension program enable younger individuals to increase their labor supply? Secondly, given that the increase in the earnings is only salient among self-employed individuals, I test whether the pension relieves or provides resources that are invested by younger individuals. For this, I test whether the pension affects the working capital for these individuals. This channel can simultaneously affect both earnings and labor supply of self-employed individuals. If the working capital increases, it means these individuals have more stock and therefore can work longer hours.

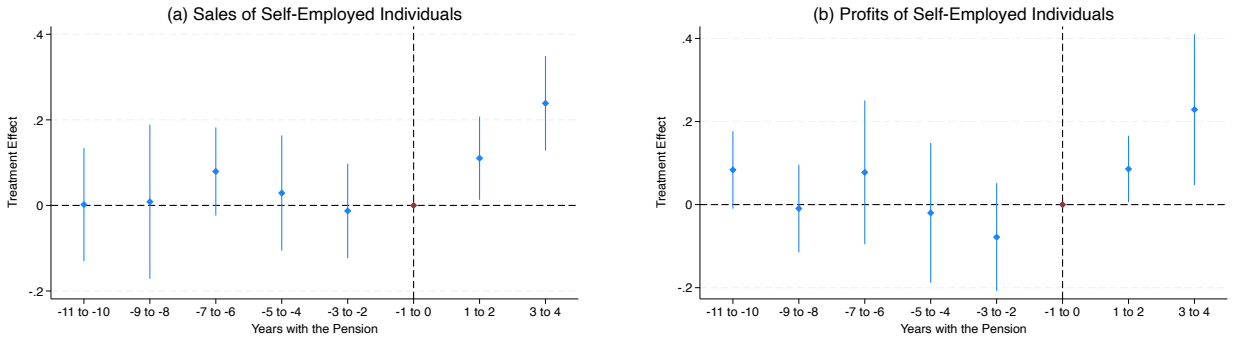



Figure 5: Old-age pension and earnings of ineligible adults in different household cohorts

*Notes:* The dependent variables are sales and profits, transformed using inverse hyperbolic sine transformation. The coefficients are estimated from regression equation 7, on a sample of individuals aged 18 to 64. The regression includes control variables (which also include interaction terms with cohorts), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level.

### *V.B. Old-Age Pension and Labor Supply of Younger Individuals*

Table 3 shows exposure to old-age pension has a positive effect on labor supply of younger individuals (aged 18 to 64 years). One additional year of pension exposure increases labor supply (weekly hours worked) by 1.3%. Based on the mean in the non-pilot, this represents an increase of 0.49 hours. This effect is also economically meaningful given the size of the pension grant as earlier highlighted. Consistent with the earnings, the effect is more significant among the self-employed individuals (1.6% increase). This suggests that the increase in earnings earlier observed is partly explained by the increase in labor supply. Among women, pension exposure increases labor supply by 1.4% overall, 1.8% among self-employed women. The effect among men is not significant although positive. Table A.9 in the Appendix shows the coefficient estimates for effects on men and women are statistically indistinguishable. I find no significant effect on labor supply among men and women in the category of employees, although the coefficients are also positive.

Similar to earnings, there might be a concern that the pension instrument is capturing effects of other features related to household composition (that might be different across pilot and non-pilot), and not only the pension. To address this concern, I follow the same approach as in section V.A, using the *placebo pension exposure* in equation 4 and estimate the regression in equation 5. Results in Table 4 show that only "true exposure" predicts positive effects of the pension on labor supply. The coefficients of the "placebo exposure" are not significant, if anything negative. This suggests that my measure of pension exposure is indeed capturing the effect of the pension.

As a further robustness check, I use a staggered difference-in-difference design to es-

Table 3: Impact of old-age pension on labor supply

| | (1) All | (2) Self-Employed | (3) Employees |
|---|---|---|---|
| *Panel A: Full Sample* | | | |
| Exposure x Pilot | 0.013* | 0.016** | 0.015 |
| | (0.007) | (0.007) | (0.012) |
| Mean in Non-Pilot | 38 | 35 | 47 |
| Observations | 16626 | 12930 | 5050 |
| $R^2$ | 0.104 | 0.098 | 0.074 |
| *Panel B: Women* | | | |
| Exposure x Pilot | 0.014** | 0.018** | 0.007 |
| | (0.007) | (0.006) | (0.019) |
| Mean in Non-Pilot | 34 | 32 | 44 |
| Observations | 8286 | 7058 | 1659 |
| $R^2$ | 0.095 | 0.085 | 0.118 |
| *Panel C: Men* | | | |
| Exposure x Pilot | 0.013 | 0.017 | 0.020 |
| | (0.012) | (0.014) | (0.016) |
| Observations | 8340 | 5872 | 3391 |
| Mean in Non-Pilot | 42 | 39 | 49 |
| $R^2$ | 0.102 | 0.108 | 0.091 |

*Notes:* In all models, the dependent variable is *ihs*(hours worked). All the models are estimated on a sample of individuals aged 18 to 64. All models include control variables (including interaction terms with PPE), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

timate the dynamic effects of the pension on labor supply. For this, I use the Uganda National Panel Survey (UNPS). Unlike earnings, the panel collects information about the hours worked for working-age individuals. Figure 6 shows that exposure to the pension program has a positive effect on hours worked, which is significant and persistent after two periods. Note that, the panel data for the period I consider is available every two years. Therefore, I interpret a period to consist of two years. Figure 6 also shows that there are no pre-trends, which suggests that the treated individuals would have been on the same trend as untreated individuals absent the pension.

I also examine the effect of the pension exposure on labor supply at the extensive margin. In Table A.10, I present the effects of the pension exposure on the likelihood of working for pay (Column 1 to 3), and being self-employed in main job (Column 4 to 6). Overall, the results show that pension exposure did not have a significant effect on the

Table 4: Placebo experiment on labor supply of individuals aged 18 to 64

| | (1) All | (2) Self-Emp | (3) Women |
|---|---|---|---|
| Exposure x Pilot | 0.013* | 0.017** | 0.014* |
| | (0.008) | (0.007) | (0.007) |
| Placebo Exposure x Pilot | -0.018 | -0.015 | -0.005 |
| | (0.013) | (0.016) | (0.013) |
| Observations | 16626 | 12930 | 8286 |
| $R^2$ | 0.107 | 0.102 | 0.098 |

*Notes:* In all models, the dependent variable is *ihs*(hours worked). All the models are estimated on a sample of individuals aged 18 to 64 years. True exposure is constructed from equation 1 while placebo exposure is constructed from equation 4. The coefficients are estimated from regression equation 5. All models include control variables (which include interaction terms between exposure measures and household/individuals characteristics), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

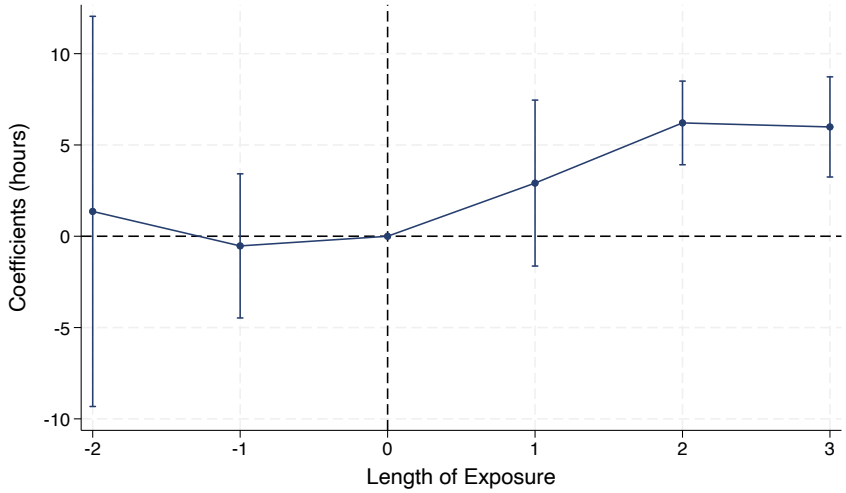



Figure 6: Dynamic effects of old-age pension on labor supply of ineligible adults

*Note:* This graph is estimated using the staggered Difference-in-Difference approach recently proposed by De Chaisemartin and d'Haultfoeuille (2020). The average treatment effect is 3.9 with standard error of 0.723. One caution to keep in mind while interpreting the graph is that the data is observed after every two years and so each period technically represents two years.

likelihood of engaging in paid employment. This result is consistent with (Huang and Zhang 2021) who show that pension exposure in China did not significantly affect the likelihood of working among the rural population. The results also show that pension exposure did not have a significant impact on the likelihood of being self-employed in the main job. Taken together, the results imply that the labor supply of ineligible adults

responds more at the intensive margin rather than the extensive margin, as a result of exposure to the old-age pension. This could be attributed to the small size of the pension grants.

The results that old-age pension increases labor supply of ineligible individuals relates to a few recent papers that have shown positive effects of cash transfers. For instance, Salehi-Isfahani and Mostafavi-Dehzooei (2018) showed that national cash transfer program in Iran had positive effects on the labor supply of women and self-employed men. More recently, Vera-Cossio (2022) showed that nationwide conditional cash transfer program in Bolivian public schools led to an increase in employment among parents of eligible children. The paper shows that the cash transfer increased the likelihood of self-employment by relaxing liquidity constraints. Even though these papers examine the effects on direct beneficiaries, they relate to my findings by showing that there are contexts in which cash transfers increase labor supply rather than decrease it, as widely cited in the literature. More closely related to my results is Ardington et al. (2009) who show that cash transfers in South Africa led to an increase in employment rates among prime-aged individuals in South Africa. They show that the pension allows prime-aged individuals to migrate for work by easing the childcare constraint. However, they only focus on the extensive margin while here, I examine the effects on both the intensive and extensive margin and show that labor supply responds more on the intensive margin. I also show that in this case, the pension did not have a significant effect on the likelihood that any household member migrates for work (Table A.17)

*Effects on the elderly:* Previous studies show that pension beneficiaries tend to alter their labor supply when they start receiving pension (de Carvalho Filho 2008; Fetter and Lockwood 2018; Unnikrishnan and Imai 2020). Although the focus of my paper is not on the effects of the pension on direct beneficiaries, I re-estimate this relationship using staggered difference-in-difference, leveraging the panel data. I find results consistent with previous studies. Figure A.1 in the Appendix shows that the elderly (above 64 years) are less likely to engage in paid employment when they start receiving pension grants. The results suggest that on average, exposure to the pension led to a 18.5 percentage point reduction in the likelihood of engaging in paid employment. This effect is larger relative to the effect reported by earlier studies in different contexts. For example, Kaushal (2014) showed that $11 (in purchasing power parity) provided through India's National Old-Age Pension Scheme was associated with a 1–3 percentage points decline in the employment of the elderly. Similarly, de Carvalho Filho (2008) showed that an increase in old-age benefits by $95 in Brazil increased the probability of not working by about 38 percentage points. The higher effect I find could be attributed to the sample size limitation

highlighted earlier, and relatively longer duration of exposure. Nonetheless, this result provides suggestive evidence that the elderly reduce their labor supply once they start receiving the pension.

Given the above evidence, I investigate whether there is reallocation of some household tasks from younger individuals to the elderly. If the elderly are exiting the labor market and are able to take on some household tasks such as caring for children, this might relieve time for younger individuals to work more. If this is happening, it could explain part of the increase in labor supply observed earlier. This could also enable them to migrate for work outside the household as highlighted by Ardington et al. (2009). To investigate this channel, I examine how the pension exposure affects time spent on household tasks by younger individuals (18 to 64 years). Table A.11 shows that exposure to pension does not significantly affect time spent on household tasks. This suggests that the increase in labor supply earlier observed is not coming from a reduction in household tasks but rather a reduction in leisure or simply from the slack in labor supply. This corroborates my finding that the pension does not induce household members to migrate for work.

### *V.C. Old-Age Pension and Working Capital of Younger Individuals*

To better understand what drives the increase in earnings and labor supply as a result of pension exposure, I test whether the pension program affects the working capital for self-employed individuals. As earlier mentioned, the self-employed individuals could provide an avenue for investment of the pension grants. The pension could also relieve financial resources for the younger individuals by supporting consumption expenditure in the household, and reducing financial dependence of the elderly. If these channels are present, we might expect the younger individuals to have more financial resources to invest in their businesses.

Table 5 shows that the pension exposure increases working capital for self-employed ineligible adults. One additional year of pension exposure increases working capital by 6.4%, which is equivalent to $2.3 increase, based on the mean in non-pilot districts. While the effect is positive for both men and women, it is insignificant among men. In a placebo experiment presented in Table A.12, I show that the effects on working capital are indeed driven by the "True Exposure" and not the "Placebo Exposure". The results on working capital are consistent with results on earnings and labor supply which show that the effects are only significant among women. The positive effect on working capital is in line with Vera-Cossio (2022) who shows that cash transfers can relax liquidity constraints and boost entrepreneurship, although Vera-Cossio (2022) studies effects on direct beneficiaries

rather than spillover. I interpret the increase in working capital to be driving the results on labor supply through the complementarity channel, and earnings. More resources can induce self-employed individuals to work longer but also has a direct effect on sales and profits.

Table 5: Impact of old-age pension on working capital for self-employed individuals

| | All | | Women | | Men | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Exposure x Pilot | 0.085** | 0.064* | 0.101** | 0.097** | 0.077 | 0.039 |
| | (0.036) | (0.034) | (0.045) | (0.044) | (0.067) | (0.062) |
| Additional Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Interview Month FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Location FE | No | Yes | No | Yes | No | Yes |
| Mean in Non-Pilot (USD) | 36.6 | 36.6 | 35.4 | 35.4 | 37.6 | 37.6 |
| Observations | 12260 | 12260 | 6885 | 6885 | 5375 | 5375 |
| $R^2$ | 0.129 | 0.204 | 0.148 | 0.234 | 0.125 | 0.188 |

*Notes:* The dependent variable is working capital (proxied by investment in raw materials/inputs), transformed using inverse hyperbolic sine transformation. All the models are estimated on a sample of individuals aged 18 to 64 years. All models include control variables (including interaction terms with potential exposure measure), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

### *V.D. Heterogeneity Analysis*

*Availability of microfinance and the effects of pension on earnings and working capital:* I explore whether the effect of the pension is different depending on the availability of micro credit in the community. In particular, I ask the question: Do pension grants compensate for limited availability of and access to micro credit? or instead, does availability of micro credit bolster the effects of the pension? If the pension grants are compensating for lack of micro credit, we would expect to see higher effects in areas with limited access to micro credit. To answer these questions, I use triple difference-in-difference design estimating the regression equation specified in equation 8.

$$\begin{aligned} y_{ihd} =& \beta_0 + \beta_1 \left( PPE_h \cdot \mathbb{I}\{Pilot_d\} \cdot \mathbb{I}\{\text{MFI} \leq 10\text{km}\} \right) + \beta_2 \left( PPE_h \cdot \mathbb{I}\{Pilot_d\} \cdot \mathbb{I}\{\text{MFI} > 10\text{km}\} \right) \\ &+ \beta_3 PPE_h + \beta_4 PPE_h^p + \beta_5 Pilot_d + \eta\prime\, Z_{hd} + \gamma'\, X_{ihd} + \tau_m + \tau_r + e_{ihd}, \end{aligned} \tag{8}$$

where $\mathbb{I}\{\text{MFI} \leq 10\text{km}\}$ is an indicator for a household being located within 10km from a microfinance and $\mathbb{I}\{\text{MFI} > 10\text{km}\}$ is an indicator for a household being located more

than 10km from a microfinance. $Z_{hd}$ is a vector of double interactions for $PPE$, $Pilot$, and distance to a microfinance. The rest of the variables are as defined before. $\beta_1$ and $\beta_2$ are the coefficients of interest.

Results presented in Table 6 show that the effects of the pension on sales, profits, and working capital are bolstered by the availability of and access to microfinance. In the full sample, the results show that the effect of the pension on the sales is 13 percentage points ($\beta_1$-$\beta_2$) higher for individuals in households located within 10km from a microfinance compared to those living more than 10km (Column 1). Similarly, the effect on profits and working capital is higher by 13 and 14 percentage points respectively, for individuals located within 10km of a microfinance. Among women, the pattern is the same, with a much higher increase in working capital for those located within 10km of a microfinance. These differences are statistically significant for sales and profits in the full sample, and sales and working capital in the women's sample. Among men, exposure to old-age pension does not produce significant effects regardless of whether they are within or far from microfinance. Broadly, these results suggest that cash transfers cannot be treated as a substitute for micro credit and do not compensate for the limited availability of and access to micro credit. Rather, the results suggest that the impact of cash transfers is bolstered by the availability of micro credit. This complementary relationship can be attributed to cash transfers increasing creditworthiness of the borrowers which stimulates demand for and access to additional capital from microfinance.

Table 6: Distance to a microfinance or bank and the effects of the pension

| | Full Sample | | | Women | | | Men | | |
|---|---|---|---|---|---|---|---|---|---|
| | (1) Sales | (2) Profits | (3) W. Capital | (4) Sales | (5) Profits | (6) W. Capital | (7) Sales | (8) Profits | (9) W. Capital |
| Exposure x Pilot x $\mathbb{I}\{\text{MFI} \leq 10\text{km}\}$ | 0.203*** | 0.193*** | 0.174** | 0.297*** | 0.274*** | 0.268** | 0.040 | 0.045 | 0.035 |
| | (0.043) | (0.039) | (0.082) | (0.072) | (0.069) | (0.103) | (0.085) | (0.076) | (0.156) |
| Exposure x Pilot x $\mathbb{I}\{\text{MFI} > 10\text{km}\}$ | 0.072 | 0.065 | 0.031 | 0.137*** | 0.137*** | 0.037 | 0.035 | 0.019 | 0.039 |
| | (0.042) | (0.041) | (0.041) | (0.048) | (0.046) | (0.042) | (0.065) | (0.063) | (0.074) |
| P-values ($\beta_1 = \beta_2$) | 0.065 | 0.054 | 0.178 | 0.093 | 0.122 | 0.088 | 0.962 | 0.782 | 0.981 |
| Observations | 14609 | 14608 | 12260 | 8579 | 8579 | 6885 | 6030 | 6029 | 5375 |
| $R^2$ | 0.194 | 0.194 | 0.204 | 0.131 | 0.131 | 0.235 | 0.219 | 0.213 | 0.188 |

*Notes:* The depend variables are sales, profits, and working capital, transformed using inverse hyperbolic sine transformation. All the models are estimated on a sample of individuals aged 18 to 64. All models include control variables, double interaction terms, interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

*Do the pension grants provide informal insurance to the younger individuals?* To answer this question, I examine the heterogeneity of the effect of the pension based on the availability of informal safety nets in the community. If the pension grants are providing informal insurance to the younger individuals, this could allow them to take on high-risk but high-

return ventures which could increase their earnings. In that case, we would expect to see higher effects in areas with limited access to informal safety nets. To test this, I estimate a regression similar to equation 8, replacing distance to microfinance with the availability of informal safety nets in the community.

In Table 7, Panel A, I shows heterogeneity based on the presence of a Savings and Credit Co-operative (SACCO) in the community and Panel B shows heterogeneity based on the presence of any informal group in the community (including SACCOs, youth groups, women groups, and business associations). Results show that the effects of the pension are higher in locations that have access to informal safety nets. Panel A shows that the pension only has a significant effect on sales, profits, and working capital in communities that have access to SACCOs. In Panel B, results show that having any informal group bolsters the effect of the pension on earnings and working capital. Communities that lack any form of informal safety nets see no effect at all in the full sample. The differences in the effect of the pension are statistically significant for sales and profits in the full sample and for all the three outcomes in the women's sample. These results suggest that the pension enables these younger individuals to leverage the existing informal networks, suggesting cash transfers can be effective in enhancing productive use of the existing social networks.

Table 7: Availability of informal safety nets and the effects of the pension

| | Full Sample | | | Women | | | Men | | |
|---|---|---|---|---|---|---|---|---|---|
| | (1) Sales | (2) Profits | (3) W. Capital | (4) Sales | (5) Profits | (6) W. Capital | (7) Sales | (8) Profits | (9) W. Capital |
| *Panel A: SACCO in the Community* | | | | | | | | | |
| Exposure x Pilot x $\mathbb{I}$\{SACCO\} | 0.137*** | 0.125*** | 0.120*** | 0.241*** | 0.228*** | 0.181** | 0.090* | 0.073 | 0.083* |
| | (0.019) | (0.020) | (0.028) | (0.051) | (0.049) | (0.067) | (0.051) | (0.053) | (0.044) |
| Exposure x Pilot x $\mathbb{I}$\{No SACCO\} | 0.046 | 0.045 | 0.010 | 0.112 | 0.107 | 0.061 | -0.004 | -0.008 | -0.012 |
| | (0.084) | (0.078) | (0.061) | (0.083) | (0.079) | (0.049) | (0.106) | (0.100) | (0.105) |
| P-values ($\beta_1 = \beta_2$) | 0.270 | 0.292 | 0.115 | 0.181 | 0.184 | 0.185 | 0.443 | 0.483 | 0.390 |
| Observations | 18914 | 18915 | 15337 | 11243 | 11243 | 8598 | 7671 | 7672 | 6739 |
| $R^2$ | 0.207 | 0.206 | 0.251 | 0.126 | 0.126 | 0.278 | 0.265 | 0.258 | 0.238 |
| *Panel B: Informal Group in the Community* | | | | | | | | | |
| Exposure x Pilot x $\mathbb{I}$\{Any Informal Group\} | 0.166*** | 0.152*** | 0.136** | 0.260*** | 0.246*** | 0.218** | 0.094** | 0.074* | 0.064* |
| | (0.034) | (0.032) | (0.053) | (0.075) | (0.071) | (0.097) | (0.039) | (0.043) | (0.035) |
| Exposure x Pilot x $\mathbb{I}$\{No Informal Group\} | -0.005 | -0.004 | -0.015 | 0.068 | 0.066 | -0.013 | -0.004 | -0.005 | 0.030 |
| | (0.048) | (0.046) | (0.062) | (0.062) | (0.062) | (0.062) | (0.082) | (0.078) | (0.113) |
| P-values ($\beta_1 = \beta_2$) | 0.001 | 0.001 | 0.150 | 0.016 | 0.018 | 0.091 | 0.236 | 0.315 | 0.772 |
| Observations | 18914 | 18915 | 15337 | 11243 | 11243 | 8598 | 7671 | 7672 | 6739 |
| $R^2$ | 0.207 | 0.206 | 0.251 | 0.126 | 0.126 | 0.279 | 0.264 | 0.256 | 0.237 |

*Notes:* The depend variables are sales, profits, and working capital, transformed using inverse hyperbolic sine transformation. All the models are estimated on a sample of individuals aged 18 to 64. All models include control variables, double interaction terms, interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

Even though this paper does not study a direct cash transfer to the analysis sample (18-64 years), the results in Table 6, and 7 suggest stronger complementarities between

cash transfers and the local financial and social landscape.

### *V.E. Does the Gender of Pension Recipient Matter?*

Earlier literature about the effects of pension programs shows that the effects might depend on the gender of the recipient. In particular Duflo (2003) showed that pensions received by women rather than men had a large impact on the anthropometric measures of girls. Similarly, Unnikrishnan and Imai (2020) shows the positive effects of old-age pensions on consumption expenditures are only observed when the recipients are women. Accordingly, I also examine whether the gender of the pension recipient matters for labor market outcomes of adult ineligible individuals. To achieve this, I construct two measures of pension exposure for each household, based on the number of eligible male and female individuals in the household as shown in equation 9.

$$FPE_h = \sum_{t=2011}^{2016} \left( \sum_{i \in h} \mathbb{I}\{Age_{iht} \geq 65\} * \mathbb{I}\{Female_{ih}\} \right), \tag{9}$$

where $FPE_h$ is female-based pension exposure, $Female_{ih}$ is an indicator variable if individual $i$ in household $h$ is female. I construct male-based exposure in an analogous way, which I label $MPE_h$. With these two measures of gender-based pension exposure, I estimate a modified version of equation 2 replacing $PPE_h$ with the two gender-based measures as shown in equation 10.

$$\begin{aligned} y_{ihd} =& \beta_0 + \beta_1(FPE_h \times Pilot_d) + \beta_2(MPE_h \times Pilot_d) + \beta_3 FPE_h + \beta_4 MPE_h \\ &+ \beta_5 Pilot_d + \gamma' X_{ihd} + \tau_m + \tau_r + e_{ihd}, \end{aligned} \tag{10}$$

all the other variables are as defined before. The coefficients of interest here are $\beta_1$ and $\beta_2$ which capture the effects of female-based and male-based pension exposure respectively.

Figure 7 presents the results for this analysis in the full sample and women's sample for outcomes on earnings. The results show that male-based exposure rather than female-based exposure has a significant effect on earnings. As shown in Figure 7a, in the full sample, one additional year of male-based pension exposure increases sales by 12.4%, profits by 10.3% (though not significant), and working capital by 7.2%. All coefficients for female-based exposure are not significant. Even though the effects of female-based and male-based exposure are statistically indistinguishable in the full sample (Table A.15), the differences are more salient in the women's sample (Figure 7b and Table A.15, Panel B). One additional year of male-based pension exposure leads to a 19.7% increase in sales, a 16.7% increase in profits, and 11.5% increase in the working capital, among women

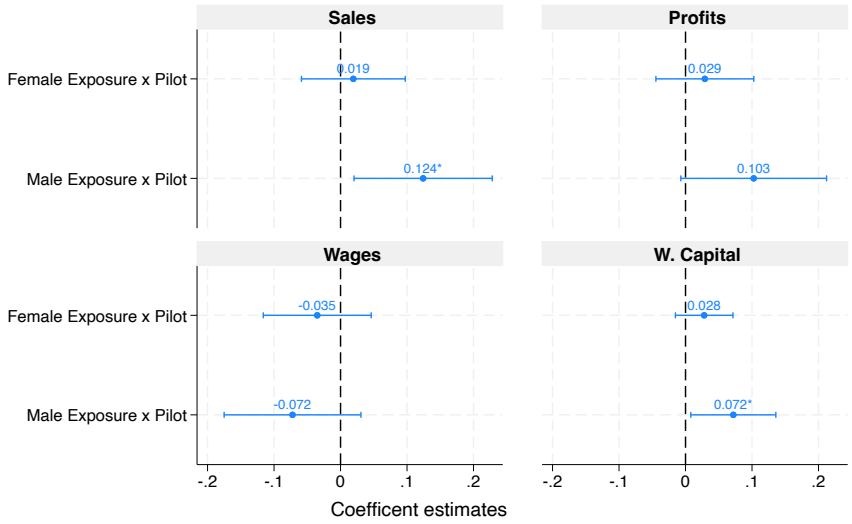
while female-based pension exposure has a mute effect. Table A.15 indeed shows that the coefficient estimates for male-based and female-based exposure are statistically different for sales and working capital (see p-values in Panel B). I however find that the gender of the recipient does not differentially influence the impact of the pension on earnings among men (Table A.15). This could be due to the weaker effects of the pension among men, to begin with.

In a similar way, I test whether the gender of the recipient matters for the effects of the pension exposure on labor supply. Results in Figure 8 show that, consistent with earnings, male-based pension exposure rather than female-based has a significant impact on labor supply. In the full sample, one additional year of male-based pension exposure increases labor supply by 2.8%, while female-based exposure has an effect of only 0.5%, which is statistically insignificant. In the women's sample, the effect of male-based pension exposure is even higher (4.8%). However, we can not reject that the effects are statistically indistinguishable (Table A.16). I also find no significant differences in the effects of male-based and female-based pension exposure in men's sample, which is consistent with the preceding evidence that the pension has small effects on men's labor market outcomes. As shown in Table A.16 in Appendix, the effect of male-based pension exposure is only significant among self-employed individuals (column 2). This is consistent with the preceding evidence that pension exposure mostly affects labor market outcomes of self-employed individuals rather than employees.

At first glance, the results that male-based rather than female-based pension exposure has a significant impact on labor market outcomes of adult ineligible individuals seems contrary to earlier evidence by Duflo (2003) and Unnikrishnan and Imai (2020). In particular, (Duflo 2003) shows that pension received by women has a large impact on children's health outcomes, especially girls, while pension received by men does not have such effects. To reconcile my results with this earlier evidence, I examine how the gender of the pension recipient affects children's outcomes in the same context that I study. For this, I use the Demographic and Health Survey for Uganda conducted in 2016. I study the effects of the pension on school attendance for primary school-going age and weight-for-height for children aged 0 to 59 months. Weight-for-height is preferred because it tends to respond faster to changes in environmental conditions (Duflo 2003). Therefore, if the pension was affecting children's health outcomes, its effects would be easily detected on weight-for-height.

Results in Table 8 show that female-based pension exposure rather than male-based exposure has a significant effect on children's outcomes. I find one additional year of female-based pension exposure increases the likelihood of attending school among pri-

(a) Full Sample




(b) Women

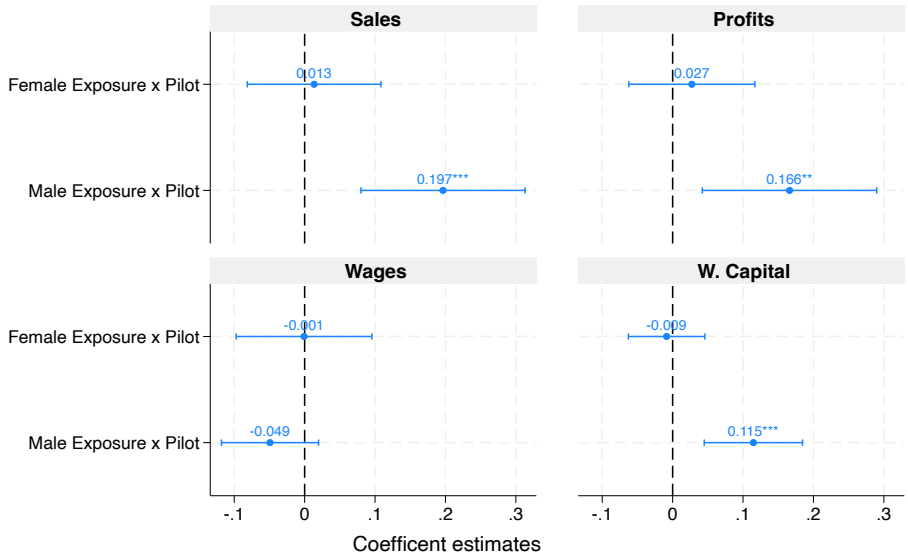



Figure 7: Gender of the recipient and effects of the pension on earnings.

*Notes:* The dependent variables are sales, profits, and working capital, transformed using IHS. Panel (a) shows effects in the full sample, while Panel (b) shows effects among women. It shows different effects for male-based exposure and female-based exposure. The dependent variable is transformed using inverse hyperbolic sine transformation. All the models are estimated on a sample of individuals aged 18 to 64 years. All models include control variables (including interaction terms with PPE), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

mary school age children by 1.6 percentage points (equivalent to 4.8% based on the mean in non-pilot districts), reduces the likelihood of being wasted by 2 percentage points

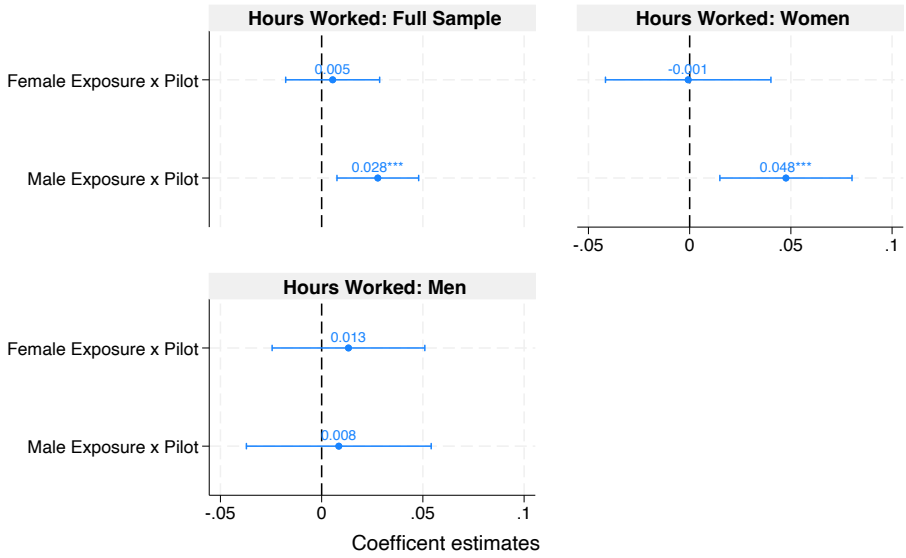



Figure 8: Gender of the recipient and effects of the pension on labor supply

*Notes:* The figure presents coefficient estimates of the effect of pension exposure on hours worked, transformed using IHS. It shows different effects for male-based exposure and female-based exposure. All the models are estimated on a sample of individuals aged 18 to 64 years. All models include control variables (including interactions with PPE), interview month fixed effects, and location fixed effects. Each plot represents a different sample as indicated in the headings. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

(equivalent to 60.6% based on the mean in non-pilot). All the estimates for male-based pension exposure are insignificant. The p-values for the test of the differences in the effects of female-based and male-based pension exposure show that the effect of female-based exposure is significantly different from the effect of male-based exposure on school attendance (column 1 and 2) and likelihood of being wasted (column 3 and 4). This result is consistent with earlier evidence that women's pension exposure rather than men's improves children's outcomes.

In summary, my results suggest that male-based pension exposure has a significant effect on labor market outcomes of ineligible adults while female-based pension exposure has a significant effect on children's outcomes. These results reflect gender differences in preferences, with men preferring to invest in income generation and women preferring to invest in children's human capital. The fact that pension received by women is largely spent on household consumption and investment in children, the most feasible channel through which it could affect earnings of younger individuals is easing financial pressure on these individuals. Given that female-based pension exposure doesn't have a stronger effect on younger individuals, this channel is likely weaker in this context. Rather, the di-

Table 8: Gender of recipient and effect of pension on children's outcomes

| | School Attendence | | Weight-for-Height | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) Wasted (1/0) | (4) Wasted (1/0) | (5) Z-Score (SD) | (6) Z-Score (SD) |
| Female Exposure x Pilot | 0.015* | 0.016** | -0.019*** | -0.020*** | 0.087 | 0.086 |
| | (0.008) | (0.008) | (0.006) | (0.007) | (0.074) | (0.075) |
| Male Exposure x Pilot | -0.003 | -0.007 | -0.000 | 0.001 | 0.063 | 0.074 |
| | (0.003) | (0.007) | (0.007) | (0.007) | (0.068) | (0.053) |
| P-values ($\beta_1 = \beta_2$) | 0.090 | 0.064 | 0.065 | 0.058 | 0.844 | 0.908 |
| Mean in Non-Pilot | 0.335 | 0.335 | 0.033 | 0.033 | 0.098 | 0.098 |
| Additional Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Interview Month FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Location FE | No | Yes | No | Yes | No | Yes |
| Observations | 4110 | 4110 | 3947 | 3947 | 3947 | 3947 |

*Notes:* School attendance is a dummy variable taking 1 if the individual of primary school going age is attending primary school, 0 otherwise. Wasted is a dummy variable taking 1 if the weight-for-height z-score is minus 2 (-2.0) standard deviations (SD) below the median on the WHO Child Growth Standards. All models include control variables (including interaction terms with PPE), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

rect investment of pension grants driven by male-based exposure seems to be the stronger channel in this case.

### *V.F. Further Robustness Checks*

Besides the robustness checks based on the placebo experiment, parallel trends plots, and the panel analysis presented earlier, I conduct the following checks. First, I test whether my results are driven by features in the capital city. To test this, I re-estimate the regressions while excluding Kampala from the sample. The results show that exclusion of Kampala from the sample does not affect the results on both earnings (Table A.5 in Appendix) and labor supply (Table A.7 in Appendix). This suggests that the results are not driven by the capital city. I also test whether the results are sensitive to the control group. For this, I first restrict the control to only districts in Phase II. I then expand the sample to include all the districts. This exercise shows that the results are not sensitive to the control groups although they tend to be slightly smaller in the magnitudes when restricting the control to Phase II districts, which is reasonable since this group was partially treated by the time of my data (see Tables A.6) and A.8.

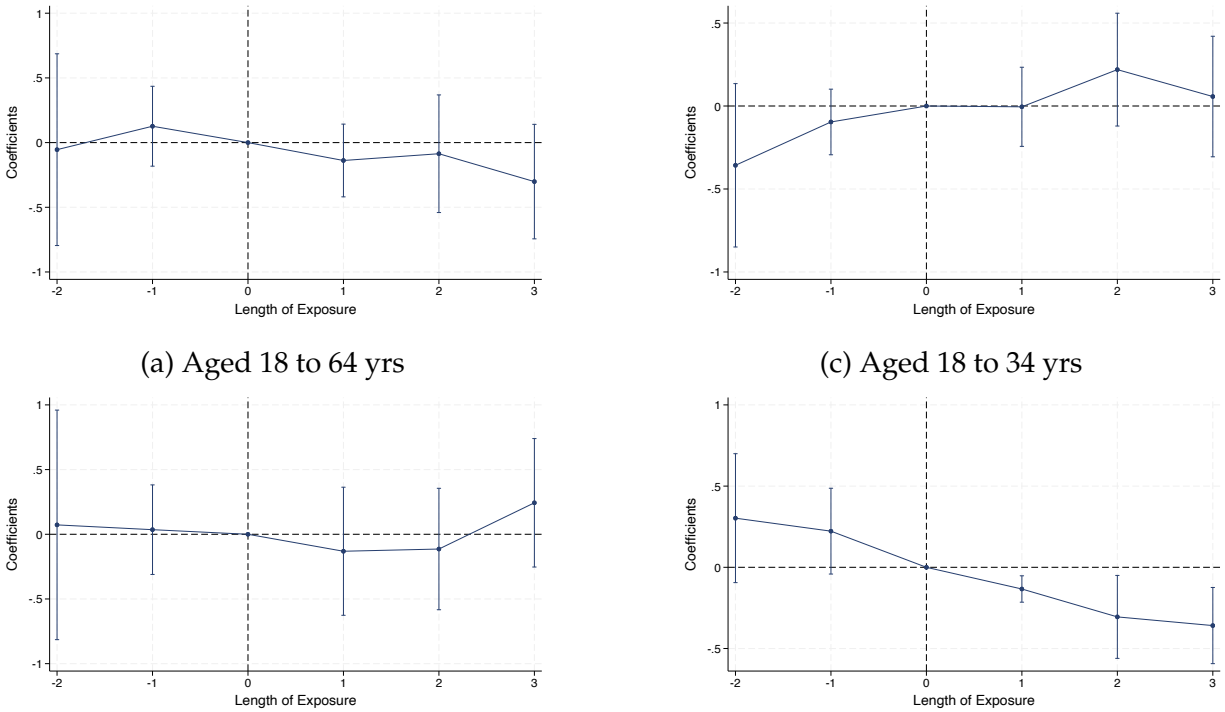
I then test whether remittances are playing a role in the results on earnings and labor supply. As shown by Fan (2010) pension grants can crowd out private transfers received by the elderly. If this is happening, my analysis might understate the effect of the pension.

First, I examine whether old-age pension impacts the amount of remittances received by the household. Table A.13 shows that indeed the pension crowds out remittances. One additional year of pension exposure reduces annual remittances received by the household by 12%. Given this result, I re-estimate the models controlling for remittance, so as to compare households with the same level of remittances. Results in Table A.14 in the Appendix show that the results are still robust when controlling for the monetary value of remittances received by the household.

### *V.G. Potential Threats to the Identification and Shortcomings*

A potential threat of my baseline specification in equation 2 is that it could be biased if the introduction of the pension program induces differential endogenous changes in household composition. I test whether this kind of endogenous formation of households is happening using the panel data. I define the outcome variable to be the number of household individuals in different age brackets at different points in time. If the pension program induces individuals to form around the pension recipients, we would expect to see different trends in treated households compared to untreated households. Evidence in Figure 9 suggests that this kind of endogenous household formation is not happening. For the age group 18 to 64 years (which is the focus of this paper), there is no differential change in the number of household members across treated and non-treated households, before and after the pension program (Figure 9a). I see a similar pattern for age groups less than 18 years (children) (Figure 9b) and 18 to 35 years (the youths) (Figure 9c). I do however see a downward trend in the number of household members aged 35 to 64 years (Figure 9d). This trend is unlikely to be driven by the pension program since there is a pre-trend before the pension program. Relatedly, if the pension induces high ability individuals to migrate for work, then studying the effects on individuals that stay in the household might understate the effects of the pension. In Table A.17 I show that the pension does not lead to migration for work.

One potential shortcoming of my analysis is that I only observe individual's exposure to the pension if they live in the same household as the pension beneficiary. If the channel through which the pension affects earnings of ineligible adults is through reduction in financial dependence by the elderly, then the effects can extend to relatives outside the households. In this study, I am not able to capture those extended effects because of data limitations. My results therefore should be interpreted as spillover effects within the household rather than broader spillovers to extended relatives.




(a) Aged 18 to 64 yrs

(c) Aged 18 to 34 yrs

(b) Below 18 yrs

(d) Aged 35 to 64 yrs

Figure 9: Dynamic effects of pension on the number of household members

*Note:* These plots are estimated using the staggered Difference-in-Difference approach recently proposed by De Chaisemartin and d'Haultfoeuille (2020). They show the trends in the number of household members in the age range indicated. One caution to keep in mind while interpreting the graph is that the data is observed after every two years and so each period technically represents two years.

## VI. CONCLUSION

This paper examines how old-age pension affects the labor market outcomes of ineligible adults within the household. While many countries have non-contributory old-age pension, the effects on ineligible individuals have been understudied. Prior work has shown spillover effects on children. In this paper, I rationalize the idea that old-age pension might affect resource availability (both financial and time) for younger individuals. For instance, by reducing financial dependence and supporting consumption expenditures, the pension grants might relieve financial resources for younger individuals. In addition, the younger individuals operating small businesses might provide avenues for investment of the pension grants. Finally, if the pension leads the elderly to exit the labor market as prior research has shown, there might be re-allocation of some household tasks to the elderly which might relieve some time for the younger individuals. Through a combination of these forces, old-age pension might have an impact on labor market

outcomes of younger individuals.

To address the question of interest, I exploit the quasi-experimental variation in the roll-out of the old-age pension program in Uganda—the Senior Citizen Grant. I construct an instrument for pension exposure based on household age composition and use difference-in-difference design to estimate the causal effects on labor market outcomes of individuals aged 18 to 64 years. I show that the pension exposure has significant positive effects on labor market outcomes of self-employed individuals, more so among women. I find evidence consistent with the idea that old-age pension relaxes the financial resource constraint of younger individuals. In this case, I show that exposure to old-age pension increases working capital of self-employed individuals, leading to higher earnings and labor supply. I show that the impact of pension exposure is bolstered by the availability of micro credit and stronger informal networks, suggesting complementarities between cash transfers and local financial and social landscape. The results are stronger in urban areas, suggesting that the effects of the pension on earnings are strengthened by other business opportunities that are associated with urban areas. I also find evidence that male-based pension exposure rather than female-based has a significant effect on labor market outcomes of ineligible adults. This is contrary to the results on children's outcomes where female-based exposure has a significant effect. To the extent that the effects on earnings are driven by investment of the pension grants by younger individuals, the difference in these results is a reflection of gender preferences, with women preferring to invest in children's human capital and men preferring to invest in income generation. I show that my results are not driven by endogenous household formation, migration, or pre-existing trends. The results are also robust to exclusion of Kampala (the capital city) from the sample, and controlling for remittances received by the household. Taken together, these results suggest that old-age pension programs have a developmental impact even on individuals that are ineligible.

REFERENCES


Akee, R. K., Copeland, W. E., Keeler, G., Angold, A. and Costello, E. J. (2010), 'Parents' incomes and children's outcomes: a quasi-experiment using transfer payments from casino profits', *American Economic Journal: Applied Economics* **2**(1), 86–115.

Alzua, M. L., Cantet, N., Dammert, A. C. and Olajide, D. (2023), 'The well-being effects of an old-age pension: Experimental evidence for ekiti state in nigeria', *Journal of African Economies* p. ejad012.

Ardington, C., Case, A. and Hosegood, V. (2009), 'Labor supply responses to large social transfers: Longitudinal evidence from south africa', *American economic journal: Applied economics* **1**(1), 22–48.

Baird, S., Ferreira, F. H., Özler, B. and Woolcock, M. (2014), 'Conditional, unconditional and everything in between: a systematic review of the effects of cash transfer programmes on schooling outcomes', *Journal of Development Effectiveness* **6**(1), 1–43.

Bukuluki, P. and Watson, C. (2012), 'Transforming cash transfers: Beneficiary and community perspectives on the senior citizen grant (scg) in uganda', *Overseas Development Institute* .

de Carvalho Filho, I. E. (2008), 'Old-age benefits and retirement decisions of rural elderly in brazil', *Journal of Development Economics* **86**(1), 129–146.

De Chaisemartin, C. and d'Haultfoeuille, X. (2020), 'Two-way fixed effects estimators with heterogeneous treatment effects', *American Economic Review* **110**(9), 2964–2996.

Duflo, E. (2003), 'Grandmothers and granddaughters: old-age pensions and intrahousehold allocation in south africa', *The World Bank Economic Review* **17**(1), 1–25.

Fan, E. (2010), 'Who benefits from public old age pensions? evidence from a targeted program', *Economic Development and Cultural Change* **58**(2), 297–322.

Fetter, D. K. and Lockwood, L. M. (2018), 'Government old-age support and labor supply: Evidence from the old age assistance program', *American Economic Review* **108**(8), 2174–2211.

Harasty, C. and Ostermeier, M. (2020), *Population Ageing: Alternative measures of dependency and implications for the future of work*, number 5, ILO Working Paper.

Haushofer, J. and Shapiro, J. (2016), 'The short-term impact of unconditional cash transfers to the poor: experimental evidence from kenya', *The Quarterly Journal of Economics* **131**(4), 1973–2042.

Huang, W. and Zhang, C. (2021), 'The power of social pensions: Evidence from china's new rural pension scheme', *American Economic Journal: Applied Economics* **13**(2), 179–205.

Kaushal, N. (2014), 'How public pension affects elderly labor supply and well-being: Evidence from india', *World Development* **56**, 214–225.

Miglino, E., Navarrete H, N., Navarrete H, G. and Navarrete H, P. (2023), 'Health effects of increasing income for the elderly: evidence from a chilean pension program', *American Economic Journal: Economic Policy* **15**(1), 370–393.

Moscona, J. and Seck, A. A. (2024), 'Age set versus kin: Culture and financial ties in east africa', *American Economic Review* **114**(9), 2748–2791.

Novignon, J., Prencipe, L., Molotsky, A., Valli, E., de Groot, R., Adamba, C. and Palermo, T. (2022), 'The impact of unconditional cash transfers on morbidity and health-seeking behaviour in africa: evidence from ghana, malawi, zambia and zimbabwe', *Health Policy and Planning* **37**(5), 607–623.

Salehi-Isfahani, D. and Mostafavi-Dehzooei, M. H. (2018), 'Cash transfers and labor supply: Evidence from a large-scale program in iran', *Journal of Development Economics* **135**, 349–367.

Shah, H. and Gennetian, L. A. (2023), 'Unconditional cash transfers for families with children in the us: A scoping review'.

Unnikrishnan, V. and Imai, K. S. (2020), 'Does the old-age pension scheme improve household welfare? evidence from india', *World Development* **134**, 105017.

Vera-Cossio, D. A. (2022), 'Dependence or constraints? cash transfers and labor supply', *Economic Development and Cultural Change* **70**(4), 1439–1477.

Verlaat, T., Todeschini, F. and Ramos, X. (2023), 'The employment effects of generous and unconditional cash support', *Available at SSRN 4390260* .

# A. APPENDIX

## *A.A. Additional Tables and Figures*

Table A.1: Differences in means for pilot and non-pilot districts in 2011, clustered

| | Mean in Non-Pilot | Diff. (Pilot-Non-Pilot) | $N$ |
|---|---|---|---|
| Potential Pension Exposure | 0.224 | 0.050 | 11215 |
| | (0.014) | (0.052) | |
| Hours worked (weekly) | 24.752 | -3.586 | 6193 |
| | (1.602) | (2.036) | |
| Self-employed in main job | 0.113 | -0.013 | 6989 |
| | (0.017) | (0.033) | |
| Self-employed in secondary job | 0.350 | 0.081 | 900 |
| | (0.018) | (0.048) | |
| Paid work | 0.136 | -0.005 | 11215 |
| | (0.012) | (0.022) | |
| Running a business | 0.099 | 0.032 | 11215 |
| | (0.007) | (0.016) | |
| Monthly wage (USD) | 88.623 | -26.644* | 546 |
| | (5.730) | (11.894) | |
| Female | 0.514 | 0.003 | 11215 |
| | (0.004) | (0.010) | |
| Age | 24.389 | 0.217 | 11183 |
| | (0.175) | (0.510) | |
| Years of schooling | 5.028 | -0.773 | 9724 |
| | (0.228) | (0.392) | |

Note: This table presents mean differences between pilot and non-pilot (excl. Phase II) districts. Each row is a regression of the named variable on a constant and the indicator for pilot. Standard errors are clustered at sub-county level. Standard errors are presented in parenthesis. *** p<0.01, ** p< 0.05, * p<0.10.

Table A.2: Impact of old-age pension on earnings

| | (1) *ihs*(Sales) | (2) *ihs*(Profits) | (3) *ihs*(Wages) |
|---|---|---|---|
| *Panel A: Full Sample* | | | |
| Exposure x Pilot | 0.102*** | 0.095*** | -0.030 |
| | (0.034) | (0.034) | (0.027) |
| Mean in Non-Pilot (USD) | 115 | 51 | 95 |
| Observations | 14609 | 14608 | 16793 |
| $R^2$ | 0.194 | 0.194 | 0.212 |
| *Panel B: Women* | | | |
| Exposure x Pilot | 0.180*** | 0.174*** | 0.017 |
| | (0.046) | (0.045) | (0.024) |
| Mean in Non-Pilot (USD) | 93 | 36 | 74 |
| Observations | 8579 | 8579 | 9376 |
| $R^2$ | 0.131 | 0.131 | 0.200 |
| *Panel C: Men* | | | |
| Exposure x Pilot | 0.036 | 0.024 | -0.070 |
| | (0.056) | (0.055) | (0.041) |
| Observations | 6030 | 6029 | 7417 |
| Mean in Non-Pilot (USD) | 136 | 64 | 109 |
| $R^2$ | 0.219 | 0.213 | 0.214 |

*Notes:* The dependent variables are sales, profits, and wages, transformed using Inverse Hyperbolic Sine Transformation (IHS). All the models are estimated on a sample of individuals aged 18 to 64. All models include control variables (including interaction terms with PPE), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level *** p<0.01, ** p< 0.05, * p<0.10.

Table A.3: Testing for differences in the effect of pension on earnings among men and women

| | Full Sample | | | Excl. Kampala | | |
|---|---|---|---|---|---|---|
| | (1) *ihs*(sales) | (2) *ihs*(profits) | (3) *ihs*(wages) | (4) *ihs*(sales) | (5) *ihs*(profits) | (6) *ihs*(wages) |
| Exposure x Pilot x Female | 0.183*** | 0.178*** | 0.038 | 0.181*** | 0.177*** | 0.033 |
| | (0.038) | (0.038) | (0.024) | (0.038) | (0.037) | (0.024) |
| Exposure x Pilot x Male | 0.036 | 0.024 | -0.096* | 0.036 | 0.023 | -0.097* |
| | (0.058) | (0.055) | (0.050) | (0.059) | (0.055) | (0.050) |
| P-values ($\beta_1 = \beta_2$) | 0.079 | 0.051 | 0.011 | 0.083 | 0.053 | 0.015 |
| Observations | 14609 | 14608 | 16995 | 13843 | 13842 | 15781 |
| $R^2$ | 0.195 | 0.195 | 0.212 | 0.191 | 0.191 | 0.207 |

*Notes:* The dependent variables are sales, profits, and wage, transformed using inverse hyperbolic since transformation. All the models are estimated on a sample of individuals aged 18 to 64. All models include control variables (including interaction terms with PPE), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

Table A.4: Placebo experiment on Earnings: Based on 55 to 64 years

| | *ihs*(Sales) | | *ihs*(Profits) | |
|---|---|---|---|---|
| | (1) All | (2) Women | (3) All | (4) Women |
| Exposure x Pilot | 0.099** | 0.179*** | 0.091** | 0.171*** |
| | (0.038) | (0.059) | (0.037) | (0.056) |
| Placebo Exposure x Pilot | 0.018 | 0.040 | 0.027 | 0.030 |
| | (0.097) | (0.134) | (0.096) | (0.132) |
| Observations | 18914 | 11243 | 18915 | 11243 |
| $R^2$ | 0.200 | 0.113 | 0.200 | 0.112 |

*Notes:* The dependent variables are *ihs*(sales) and *ihs*(profits). All the models are estimated on a sample of individuals aged 18 to 64 years. True exposure is constructed from equation 1 while placebo exposure is constructed from equation 4 by adjusting the lower threshold to 55. The coefficients are estimated from equation 5. All models include control variables (including interaction terms with PPE), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

Table A.5: Impact of the pension on earnings (Excl. Kampala)

| | (1) *ihs*(Sales) | (2) *ihs*(Profits) | (3) *ihs*(Wages) |
|---|---|---|---|
| *Panel A: Full Sample* | | | |
| Exposure x Pilot | 0.102*** | 0.094*** | -0.034 |
| | (0.034) | (0.033) | (0.026) |
| Mean in Non-Pilot (USD) | 107 | 48 | 89 |
| Observations | 13843 | 13842 | 15598 |
| $R^2$ | 0.190 | 0.190 | 0.207 |
| *Panel B: Women* | | | |
| Exposure x Pilot | 0.176*** | 0.170*** | 0.013 |
| | (0.046) | (0.045) | (0.023) |
| Mean in Non-Pilot (USD) | 86 | 34 | 68 |
| Observations | 8075 | 8075 | 8683 |
| $R^2$ | 0.129 | 0.128 | 0.198 |
| *Panel C: Men* | | | |
| Exposure x Pilot | 0.035 | 0.023 | -0.076* |
| | (0.058) | (0.056) | (0.040) |
| Observations | 5768 | 5767 | 6915 |
| Mean in Non-Pilot (USD) | 126 | 61 | 102 |
| $R^2$ | 0.209 | 0.204 | 0.210 |

*Notes:* The dependent variables are indicated in the column headings. These are transformed using inverse hyperbolic sine transformation. All the models are estimated on a sample of individuals aged 18 to 64. All models include control variables (including interaction terms with PPE), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

Table A.6: Impact of the pension on earnings: Alternative control groups

| | Only Phase II as control | | | Including all districts | | |
|---|---|---|---|---|---|---|
| | (1) *ihs*(Sales) | (2) *ihs*(Profits) | (3) *ihs*(Wages) | (4) *ihs*(Sales) | (5) *ihs*(Profits) | (6) *ihs*(Wages) |
| *Panel A: Full Sample* | | | | | | |
| Exposure x Pilot | 0.081** | 0.078** | -0.057** | 0.098*** | 0.092*** | -0.038 |
| | (0.033) | (0.033) | (0.026) | (0.028) | (0.028) | (0.025) |
| Observations | 5868 | 5867 | 6358 | 18589 | 18588 | 21207 |
| $R^2$ | 0.180 | 0.179 | 0.184 | 0.191 | 0.190 | 0.210 |
| *Panel B: Women* | | | | | | |
| Exposure x Pilot | 0.168** | 0.166** | -0.000 | 0.175*** | 0.170*** | 0.013 |
| | (0.061) | (0.060) | (0.036) | (0.044) | (0.043) | (0.024) |
| Observations | 3254 | 3254 | 3372 | 10792 | 10792 | 11717 |
| $R^2$ | 0.133 | 0.133 | 0.142 | 0.130 | 0.130 | 0.193 |
| *Panel B: Men* | | | | | | |
| Exposure x Pilot | 0.029 | 0.021 | -0.116*** | 0.038 | 0.026 | -0.089** |
| | (0.064) | (0.060) | (0.036) | (0.055) | (0.053) | (0.037) |
| Observations | 2614 | 2613 | 2986 | 7797 | 7796 | 9490 |
| $R^2$ | 0.197 | 0.196 | 0.193 | 0.210 | 0.205 | 0.213 |

*Notes:* The dependent variables are indicated in the column headings. These are transformed using IHS. All the models are estimated on a sample of individuals aged 18 to 64. All models include control variables (including interaction terms with PPE), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

Table A.7: Impact of old-age pension on labor supply: Excluding Kampala

| | (1) All | (2) Self-Employed | (3) Employees |
|---|---|---|---|
| *Panel A: Full Sample* | | | |
| Exposure x Pilot | 0.013* | 0.016** | 0.018 |
| | (0.007) | (0.007) | (0.012) |
| Mean in Non-Pilot | 37 | 34 | 46 |
| Observations | 15654 | 12523 | 4460 |
| $R^2$ | 0.097 | 0.092 | 0.074 |
| *Panel B: Women* | | | |
| Exposure x Pilot | 0.014** | 0.018*** | 0.008 |
| | (0.006) | (0.006) | (0.021) |
| Mean in Non-Pilot | 33 | 31 | 42 |
| Observations | 7839 | 6846 | 1415 |
| $R^2$ | 0.085 | 0.079 | 0.123 |
| *Panel C: Men* | | | |
| Exposure x Pilot | 0.013 | 0.017 | 0.023 |
| | (0.012) | (0.014) | (0.017) |
| Observations | 7815 | 5677 | 3045 |
| Mean in Non-Pilot | 41 | 38 | 48 |
| $R^2$ | 0.095 | 0.101 | 0.090 |

*Notes:* In all models, the dependent variable is *ihs*(hours worked). All the models are estimated on a sample of individuals aged 18 to 64. All models include control variables (including interaction terms with PPE), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

Table A.8: Impact of old-age pension on labor supply: All districts

| | (1) All | (2) Self-Employed | (3) Employees |
|---|---|---|---|
| *Panel A: Full Sample* | | | |
| Exposure x Pilot | 0.009 | 0.013** | 0.015 |
| | (0.006) | (0.006) | (0.010) |
| Mean in Non-Pilot | 37 | 34 | 46 |
| Observations | 21025 | 16589 | 6169 |
| $R^2$ | 0.096 | 0.091 | 0.067 |
| *Panel B: Women* | | | |
| Exposure x Pilot | 0.012 | 0.015** | 0.007 |
| | (0.007) | (0.007) | (0.017) |
| Mean in Non-Pilot | 33 | 31 | 43 |
| Observations | 10438 | 8998 | 2001 |
| $R^2$ | 0.090 | 0.081 | 0.113 |
| *Panel C: Men* | | | |
| Exposure x Pilot | 0.008 | 0.012 | 0.016 |
| | (0.012) | (0.013) | (0.015) |
| Observations | 10587 | 7591 | 4168 |
| Mean in Non-Pilot | 41 | 38 | 48 |
| $R^2$ | 0.088 | 0.092 | 0.077 |

*Notes:* In all models, the dependent variable is *ihs*(hours worked). All the models are estimated on a sample of individuals aged 18 to 64. All models include control variables (including interaction terms with PPE), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

Table A.9: Testing for differences in the effect of pension on labor supply among men and women

| | Full Sample | | | Excl. Kampala | | |
|---|---|---|---|---|---|---|
| | (1) All Categories | (2) Self-Employed | (3) Employee | (4) All Categories | (5) Self-Employed | (6) Employee |
| Exposure x Pilot x Female | 0.019** | 0.021*** | 0.010 | 0.020** | 0.021*** | 0.013 |
| | (0.007) | (0.007) | (0.017) | (0.007) | (0.007) | (0.017) |
| Exposure x Pilot x Male | 0.005 | 0.011 | 0.013 | 0.005 | 0.011 | 0.015 |
| | (0.014) | (0.015) | (0.014) | (0.015) | (0.015) | (0.015) |
| P-values ($\beta_1 = \beta_2$) | 0.446 | 0.613 | 0.892 | 0.436 | 0.595 | 0.933 |
| Observations | 16626 | 12930 | 5050 | 15654 | 12523 | 4460 |
| $R^2$ | 0.105 | 0.099 | 0.075 | 0.097 | 0.092 | 0.074 |

*Notes:* In all models, the dependent variable is *ihs*(hours worked). The coefficients in the first row indicate the effects of the pension on women. The coefficients in second row are the effects on men. All the models are estimated on a sample of individuals age 18 to 64. All models include control variables (including interaction terms with PPE), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

Table A.10: Effects of old-age pension on likelihood of working for pay and being self-employed

| | Working for pay | | | Self-Employed in main job | | |
|---|---|---|---|---|---|---|
| | (1) Full Sample | (2) Women | (3) Men | (4) Full Sample | (5) Women | (6) Men |
| Exposure x Pilot | -0.000 | 0.002 | -0.001 | 0.006 | 0.008 | 0.000 |
| | (0.003) | (0.006) | (0.005) | (0.005) | (0.006) | (0.006) |
| Additional Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Interview Month FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Location FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean in Non-Pilot | 0.431 | 0.342 | 0.534 | 0.455 | 0.426 | 0.485 |
| Observations | 25663 | 13548 | 12115 | 21828 | 10771 | 11057 |
| $R^2$ | 0.173 | 0.109 | 0.175 | 0.126 | 0.114 | 0.172 |

*Notes:* The dependent variable is a dummy variable taking 1 if the individual is working for pay or self-employed, 0 otherwise. All the models are estimated on a sample of individuals aged 18 to 64 years. All models include control variables (including interaction terms with PPE), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

Table A.11: Effects of old-age pension on time spent on household task

| | (1) All | (2) Women | (3) Men |
|---|---|---|---|
| Exposure x Pilot | 0.008 | 0.009 | 0.001 |
| | (0.007) | (0.009) | (0.012) |
| Additional Controls | Yes | Yes | Yes |
| Interview Month FE | Yes | Yes | Yes |
| Location FE | Yes | Yes | Yes |
| Mean in Non-Pilot | 14 | 20 | 7 |
| Observations | 25661 | 13548 | 12113 |
| $R^2$ | 0.319 | 0.048 | 0.128 |

Note: In all the models,the dependent variable is *ihs*(hours). All the models are estimated on a sample of individuals aged 18 to 64 years. All models include control variables (including interaction terms with PPE), month of interview and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

Table A.12: Placebo on Working Capital

| | (1) All | (2) Women | (3) Men |
|---|---|---|---|
| Exposure x Pilot | 0.063* | 0.091* | 0.037 |
| | (0.034) | (0.045) | (0.064) |
| Placebo Exposure x Pilot | -0.032 | 0.040 | -0.089* |
| | (0.050) | (0.080) | (0.044) |
| Additional Controls | Yes | Yes | Yes |
| Interview Month FE | Yes | Yes | Yes |
| Location FE | Yes | Yes | Yes |
| Mean in Non-Pilot (USD) | 33.2 | 33.0 | 33.4 |
| Observations | 12260 | 6885 | 5375 |
| $R^2$ | 0.213 | 0.243 | 0.199 |

*Notes:* The dependent variable is working capital, transformed using IHS. All the models are estimated on a sample of individuals aged 18 to 64 years. True exposure is constructed from equation 1 while placebo exposure is constructed from equation 4. The coefficients are estimated from regression equation 5. All models include control variables (including interaction terms with PPE), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

Table A.13: Impact of old-age pension on remittances received by the household

| | (1) | (2) |
|---|---|---|
| Exposure x Pilot | -0.125** | -0.117** |
| | (0.052) | (0.050) |
| Additional Controls | Yes | Yes |
| Interview Month FE | Yes | Yes |
| Location FE | No | Yes |
| Mean in Non-Pilot (USD) | 69 | 69 |
| Observations | 12168 | 12168 |
| $R^2$ | 0.070 | 0.097 |

*Notes:* The dependent variable is remittance transformed using Inverse Hyperbolic Sine transformation. All the models are estimated on a sample of individuals aged 18 to 64. All models include control variables (including interaction terms with PPE), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

Table A.14: Old-age pension and labor market outcomes: Controlling for remittances

| | Earnings | | | Hours worked | | |
|---|---|---|---|---|---|---|
| | (1) *ihs*(Sales) | (2) *ihs*(Profits) | (3) *ihs*(Wage) | (4) All | (5) Self-Employed | (6) Employees |
| *Panel A: Full Sample* | | | | | | |
| Exposure x Pilot | 0.102*** | 0.095*** | -0.030 | 0.013* | 0.016** | 0.016 |
| | (0.034) | (0.034) | (0.026) | (0.007) | (0.007) | (0.012) |
| Observations | 14609 | 14608 | 16793 | 16626 | 12930 | 5050 |
| $R^2$ | 0.194 | 0.194 | 0.212 | 0.104 | 0.099 | 0.075 |
| *Panel B: Women* | | | | | | |
| Exposure x Pilot | 0.180*** | 0.174*** | 0.017 | 0.014** | 0.018*** | 0.009 |
| | (0.046) | (0.045) | (0.024) | (0.007) | (0.006) | (0.020) |
| Observations | 8579 | 8579 | 9376 | 8286 | 7058 | 1659 |
| $R^2$ | 0.131 | 0.131 | 0.200 | 0.095 | 0.085 | 0.119 |
| *Panel C: Men* | | | | | | |
| Exposure x Pilot | 0.033 | 0.021 | -0.069* | 0.013 | 0.017 | 0.020 |
| | (0.053) | (0.052) | (0.040) | (0.012) | (0.014) | (0.016) |
| Observations | 6030 | 6029 | 7417 | 8340 | 5872 | 3391 |
| $R^2$ | 0.220 | 0.213 | 0.214 | 0.102 | 0.109 | 0.092 |

*Notes:* In columns 1 to 3, the dependent variables are *ihs*(sales), *ihs*(profits) and *ihs*(wages). In columns 4 to 6, the dependent variable is *ihs*(hours worked). All the models are estimated on a sample of individuals aged 18 to 64. All models include control variables (including interaction terms with PPE), interview month fixed effects, and location fixed effects. In this particular table, I also control for the monetary value of remittances received by the households. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

Table A.15: Gender of the recipient and effect of pension on earnings and working capital

| | (1) *ihs*(Sales) | (2) *ihs*(Profits) | (3) *ihs*(Wages) | (4) *ihs*(W. Capital) |
|---|---|---|---|---|
| *Panel A: Full Sample* | | | | |
| Female-Based Exposure x Pilot | 0.019 | 0.029 | -0.035 | 0.028 |
| | (0.047) | (0.044) | (0.049) | (0.026) |
| Male-Based Exposure x Pilot | 0.124* | 0.103 | -0.072 | 0.072* |
| | (0.063) | (0.066) | (0.062) | (0.038) |
| P-values ($\beta_1 = \beta_2$) | 0.248 | 0.418 | 0.605 | 0.389 |
| Mean in Non-Pilot (USD) | 107.6 | 48.5 | 95.9 | 45.6 |
| Observations | 14628 | 14627 | 9544 | 18696 |
| $R^2$ | 0.232 | 0.230 | 0.476 | 0.213 |
| *Panel B: Women* | | | | |
| Female-Based Exposure x Pilot | 0.013 | 0.027 | -0.001 | -0.009 |
| | (0.057) | (0.054) | (0.058) | (0.033) |
| Male-Based Exposure x Pilot | 0.197*** | 0.166** | -0.049 | 0.115*** |
| | (0.070) | (0.074) | (0.041) | (0.042) |
| P-values ($\beta_1 = \beta_2$) | 0.068 | 0.150 | 0.516 | 0.045 |
| Mean in Non-Pilot (USD) | 87.6 | 35.3 | 78.9 | 41.5 |
| Observations | 8595 | 8595 | 6111 | 11045 |
| $R^2$ | 0.151 | 0.148 | 0.334 | 0.152 |
| *Panel C: Men* | | | | |
| Female-Based Exposure x Pilot | 0.034 | 0.040 | -0.043 | 0.065 |
| | (0.063) | (0.059) | (0.078) | (0.047) |
| Male-Based Exposure x Pilot | 0.060 | 0.048 | -0.063 | 0.011 |
| | (0.085) | (0.087) | (0.130) | (0.053) |
| P-values ($\beta_1 = \beta_2$) | 0.829 | 0.943 | 0.897 | 0.478 |
| Mean in Non-Pilot (USD) | 125.3 | 60.2 | 106.4 | 49.2 |
| Observations | 6033 | 6032 | 3433 | 7651 |
| $R^2$ | 0.284 | 0.275 | 0.600 | 0.295 |

*Notes:* The dependent variables are indicated in respective columns. All dependent variables are transformed using IHS. Column 4 is the working capital for self-employed individuals. The coefficients are estimated from equation 10. All the models are estimated on a sample of individuals aged 18 to 64. All models include control variables (including interaction terms with PPE), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

Table A.16: Gender of the recipient and effect of pension on labor supply

| | (1) All Categories | (2) Self-Employed | (3) Employees |
|---|---|---|---|
| *Panel A: Full Sample* | | | |
| Female Exposure x Pilot | 0.005 | 0.013 | -0.010 |
| | (0.012) | (0.011) | (0.037) |
| Male Exposure x Pilot | 0.028*** | 0.026** | 0.035 |
| | (0.010) | (0.011) | (0.025) |
| P-values ($\beta_1 = \beta_2$) | 0.176 | 0.470 | 0.403 |
| Mean in Non-Pilot | 36.8 | 33.6 | 45.7 |
| Observations | 16652 | 12944 | 5062 |
| $R^2$ | 0.094 | 0.083 | 0.076 |
| *Panel B: Women* | | | |
| Female Exposure x Pilot | -0.001 | 0.001 | 0.019 |
| | (0.021) | (0.021) | (0.021) |
| Male Exposure x Pilot | 0.048*** | 0.047*** | 0.030 |
| | (0.016) | (0.017) | (0.042) |
| P-values ($\beta_1 = \beta_2$) | 0.146 | 0.174 | 0.850 |
| Mean in Non-Pilot | 32.9 | 30.6 | 42.7 |
| Observations | 8302 | 7070 | 1663 |
| $R^2$ | 0.077 | 0.066 | 0.103 |
| *Panel C: Men* | | | |
| Female Exposure x Pilot | 0.013 | 0.032* | -0.038 |
| | (0.019) | (0.016) | (0.062) |
| Male Exposure x Pilot | 0.008 | 0.003 | 0.029 |
| | (0.023) | (0.025) | (0.034) |
| P-values ($\beta_1 = \beta_2$) | 0.896 | 0.438 | 0.455 |
| Observations | 8350 | 5874 | 3399 |
| Mean in Non-Pilot | 40.9 | 37.3 | 47.3 |
| $R^2$ | 0.100 | 0.101 | 0.087 |

*Notes:* In all models, the dependent variable is *ihs*(hours worked). The coefficients are estimated from equation 10. All the models are estimated on a sample of individuals aged 18 to 64. All models include control variables (including interaction terms with PPE), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

Table A.17: Effects of old-age pension on the likelihood of a household member migrating for work

| | (1) | (2) |
|---|---|---|
| Exposure x Pilot | 0.002 | 0.002 |
| | (0.003) | (0.003) |
| Additional Controls | Yes | Yes |
| Interview Month FE | Yes | Yes |
| Location FE | No | Yes |
| Mean in Non-Pilot | 0.065 | 0.065 |
| Observations | 12168 | 12168 |
| $R^2$ | 0.033 | 0.046 |

*Note:* The dependent variable is a dummy variable taking 1 if the household has at least one member who migrated for work, 0 otherwise. All models include control variables (which include interaction terms between PPE and household/individuals characteristics), interview month fixed effects, and location fixed effects. Standard errors are clustered at sub-county level. *** p<0.01, ** p< 0.05, * p<0.10.

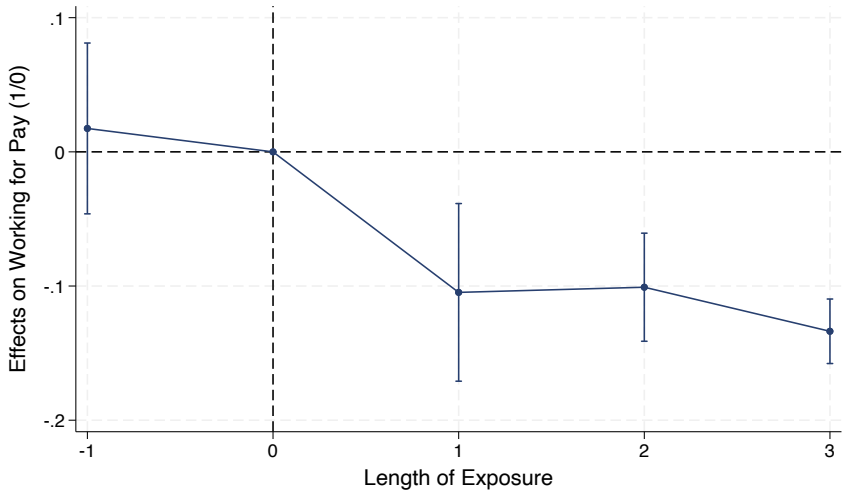



Figure A.1: Effects of old-age pension on the likelihood of working for pay: recipients

Note: This graph is estimated using the staggered Difference-in-Difference approach recently proposed by De Chaisemartin and d'Haultfoeuille (2020). The average treatment effect is -0.185 with standard error of 0.033. The graph shows how old-age pension affects the likelihood that eligible individuals are engaged in work for pay (paid employees). One caution to keep in mind while interpreting the graph is that there are gap years after the first year of treatment which are not depicted on the graph.

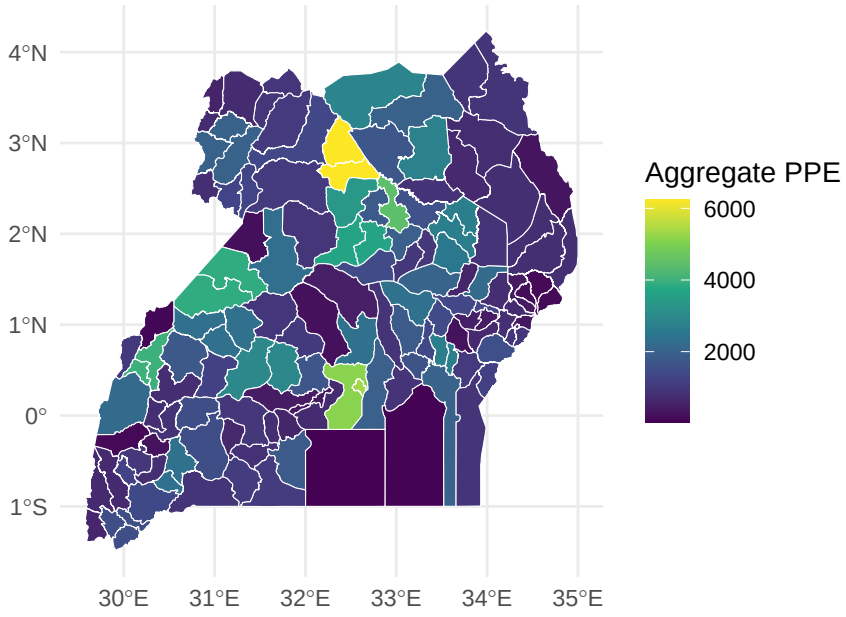



Figure A.2: **Variation in aggregate potential pension exposure by districts**

Note: This map shows variation in aggregate potential pension exposure across all the districts in Uganda. It is generated by aggregating the exposure computed using equation 1 for all the households in each district